



Bernstein's second theorem and Viro's method for sparse polynomial systems in chemistry

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Abstract

We consider a family of sparse polynomial systems defined by a directed graph and a bipartite graph which depend on certain parameters. A convex polyhedral cone serves as a representative of all positive solutions of the family. We study the boundary of this cone with Bernstein's second theorem and Viro's method. In particular we present new results about the parameter regions where several positive solutions appear. Since they are steady states of an underlying dynamical system of mass action kinetics, the resulting multistationarity has important implications for the dynamics of that system. Examples from applications illustrate the theoretical results.

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1. Introduction

We consider a family of sparse (Laurent) polynomial systems of the form

$$0 = Y_s I_a I_K \Psi(x), \quad \Psi_j(x) = x^{\eta_j}, \quad j = 1, \dots, n, \quad (1)$$

$$0 = v_i^t x - c_i, \quad i = 1, \dots, r, \quad (2)$$

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where Y_s is a fixed matrix in $(\mathbb{Z}_{\geq 0})^{m,n}$, the set of $(m \times n)$ -matrices with non-negative integer entries, I_a a fixed matrix in $\mathbb{Z}^{n,l}$, η_j is a fixed vector in $\mathbb{Z}^m$, v_i is a fixed vector in $\mathbb{R}^m$. I_K is a matrix in $(\mathbb{Q}[\underline{k}])^{l,n}$, where $\mathbb{Q}[\underline{k}]$ is the ring of polynomials in the parameters k_{ij} (which can take positive real values), and c_j are real constants. In particular, the polynomial system depends on the parameters k_{ij} and the constants c_i . These systems arise when studying the steady states of a differential system, modeling chemical reactions with *mass action kinetics*. The linear equations (2) are *conservation relations*, like, e.g., conservation of mass. There are many results about such systems in the literature in chemistry and applied mathematics, including [6,15,16,18,24,26,30,36].

For fixed values of the parameters k_{ij} the set of all positive solutions of (1) is mapped onto $\{[z_1 : z_2 : \dots : z_l] \in \mathbb{P}^{l-1} \mid z \in \ker(Y_s I_a) \cap \mathbb{R}_+^l, p_i(z; \underline{k}) = 0, i = 1, \dots, t\}$, where $p_i = z^{u_i^+} - \gamma_i(\underline{k}) z^{u_i^-} \in \mathbb{R}[z]$ are special homogeneous polynomials. The simple reason is that for a positive solution x the vector $z = I_K \Psi(x)$ is positive and in the nullspace of $Y_s I_a$. The set $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$ is a convex polyhedral cone. On the other hand z is in the image of $(\mathbb{R} \setminus \{0\})^m \rightarrow \mathbb{R}^l, x \mapsto I_K \Psi(x)$. The image is described implicitly by an ideal $I = \langle g_1, \dots, g_t \rangle \subseteq \mathbb{R}[z]$ analogous to a toric ideal. The variety $V(I) \subseteq \mathbb{P}^{l-1}$ is the Zariski closure of $\text{im}(I_K \Psi)$. It is the Zariski closure of the maximal orbit $\mathcal{O}_{I_K \mathbf{1}}$ of a torus action induced by the monomials x^{η_j} .

For each positive z in the cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$, there exists positive values of k_{ij} such that $z \in \text{im}(I_K \Psi)$ and consequently $p_i(z; \underline{k}) = 0, i = 1, \dots, t$. The set of all positive solutions of the family $\{x \in \mathbb{R}_+^m \mid \exists k_{ij} > 0 \text{ with } Y_s I_a I_K \Psi(x) = 0\}$ is mapped onto $\{[z_1 : z_2 : \dots : z_l] \in \mathbb{P}^{l-1} \mid z \in \ker(Y_s I_a) \cap \mathbb{R}_+^l\}$. Thus the interior of the convex polyhedral cone represents all positive solutions of the family. For arbitrary parameters $k_{ij} > 0$, we view $p_i = z^{u_i^+} - \gamma_i(\underline{k}) z^{u_i^-}$ as elements in $\mathbb{Q}(\underline{k})[z]$. In [21] we called $I = \langle g_1, \dots, g_t \rangle \subseteq \mathbb{Q}(\underline{k})[z]$ a *deformed toric ideal*. Background on toric geometry can be found in [7,14,19,34]. Instead of abstract toric varieties, we are using the language of embeddings of toric varieties which is more appropriate for applications.

The minimal generators of the cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$ are fundamental to the analysis of stability of steady states and the occurrence of Hopf bifurcation in [6], see also [10–12]. This is the reason why we answer the following questions in the paper: How does the boundary of the cone relate to non-negative real solutions of the family (1)? Which values of the parameters k_{ij} are relevant for this?

The main insight is that there are two different types of minimal generators of the cone. The first type are *positive circuits* E_i of a directed graph with $I_a E_i = 0$, the second type are the so-called *stoichiometric* generators ($I_a E_i \neq 0$).

In this paper we relate the positive circuits of the directed graph to Bernstein's second theorem [2,5,9,25]. This theorem is relating solutions at infinity or with zero-components to the facets of the *Newton polytopes*. A subsystem defined by a positive circuit equals under mild conditions an *initial face system*. The simple reason is that only a few chemical species are involved in each reaction or, equivalently, the monomials have sparse support. Thus a positive circuit gives asymptotic information about steady states for extreme values of mass.

Systems whose cone $\ker(Y_s, I_a) \cap (\mathbb{R}_{\geq 0})^l$ is generated by positive circuits have the property that each affine space $\{x \in \mathbb{R}^m \mid v_i^t x - c_i = 0\}$ contains a unique positive solution.

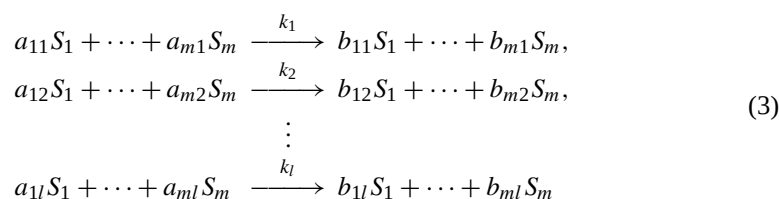
Thus stoichiometric generators are a source for several positive solutions (multistationarity). This is explained by generalizing Viro's method [9,22,32,33,35] for the study of the number of positive solutions. We replace the role of the monomials $\Psi_j(x) = x^{\eta_j}$ in the standard theory by $(I_K \Psi(x))_l = k_{ij} x^{\eta_j}$. We replace the artificial parameter in the classical method by the parameters $k_{g_i} = \underline{k}^{\bar{g}_i}$ where $g_i = I_a E_i$. An important ingredient are Stanley–Reisner ideals of *regular triangulations* of the *Cayley-graph-polytope*. Our main theorem is Theorem 4.18. Multistationarity has important implications for the dynamics of the underlying differential equation, e.g., in the context of singular perturbation theory or for the existence of reaction–diffusion waves.

2. Sparse polynomials from mass action kinetics

First we introduce the equations as presented in [20,21], being slightly more general so as to allow Laurent polynomials. Then, we outline the main principle, along with some computational issues.

2.1. Sparse equations on graphs

A system of chemical reactions with chemical species $S_1, \dots, S_m$ is given as



with given $A = (a_{ij})$, $B = (b_{ij}) \in (\mathbb{Z}_{\geq 0})^{m,l}$, and real numbers $k_i > 0$, $i = 1, \dots, l$, called *rate constants*. Then a polynomial system is defined. For each reaction a vector is formed by the differences of the coefficients on the right and the left and is multiplied with the monomial $\vec{v}_i = k_i \prod_{j=1}^m x_j^{a_{ji}}$. With $N = B - A$

$$0 = (B - A) \begin{pmatrix} k_1 x_1^{a_{11}} & \dots & x_m^{a_{m1}} \\ \vdots & & \vdots \\ k_l x_1^{a_{1l}} & \dots & x_m^{a_{ml}} \end{pmatrix} = N \vec{v}(x). \quad (4)$$

These are the equations for the steady states of the corresponding *mass action kinetics*. Sometimes the exponents differ slightly from a_{ij} . Such systems are then called *generalized mass action kinetics*.

Example 2.1. The reaction laws in Fig. 1 describe the oxidation of H_2 on a catalytic surface [10,20]. There are $m = 5$ species, and $l = 9$ reactions. $x_1, \dots, x_5$ are the concentrations of O, H, H_2O , H_2O_f and the amount of free space on the surface, respectively. The equations for the steady states of generalized mass action kinetics are here

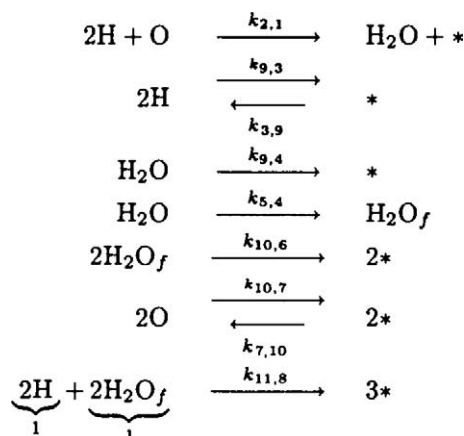


Fig. 1. Reaction laws of oxidation on catalytic surface.

$$0 = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 & -2 & 2 & 0 \\ -2 & -2 & 2 & 0 & 0 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 0 & 0 & -2 \\ 1 & 1 & -1 & 1 & 0 & 2 & 2 & -2 & 3 \end{bmatrix} \begin{bmatrix} k_{2,1}x_1x_2^2 \\ k_{9,3}x_2^2 \\ k_{3,9}x_5 \\ k_{9,4}x_3 \\ k_{5,4}x_3 \\ k_{10,6}x_4^2 \\ k_{10,7}x_1^2 \\ k_{7,10}x_5^2 \\ k_{11,8}x_2x_4 \end{bmatrix}.$$

For the last reaction, the monomial x_2x_4 instead of $x_2^2x_4^2$ has to be used as indicated by underbracing. Presenting the equations in this form has some obvious drawbacks. For example the second and third column are the same except of the sign. The monomial x_3 appears twice with different rate constants. The system will be rewritten below.

Some of the left-hand sides and right-hand sides of the reactions (3) may be equal. We refer to them as *complexes* C_j , $j = 1, \dots, n$. On this level of abstraction, (3) is a *directed graph* with vertices C_j , $j = 1, \dots, m$, and oriented edges (arrows) $C_j \rightarrow C_i$ with weights $k_{ij} > 0$. Here, C_j is called *reactant complex* or *educt complex*, while C_i is called *product complex*. This directed graph has two incidence matrices. The first incidence matrix is $I_a = (w_{ij})_{i=1,\dots,n,j=1,\dots,l}$, where $w_{ij} \in \{-1, 0, 1\}$ and l denotes the number of arrows. Each column represents an oriented edge by containing one entry -1 for the reactant and 1 for the product. The second incidence matrix is $I_K = (u_{\mu\nu})_{\mu=1,\dots,l,\nu=1,\dots,n}$ with $u_{\mu\nu} \in \{k_{ij} \mid i, j = 1, \dots, n\}$, or $u_{\mu\nu} = 0$. Each row corresponds to one oriented edge and has at most one non-zero entry. The μ th arrow $C_j \rightarrow C_i$ gives $u_{\mu j} = k_{ij}$ encoding the weight and that C_j is the initial vertex.

We still need the information that each complex C_j is a linear combination $\sum_{i=1}^m y_{ij} S_i$ of species S_i with coefficients $y_{ij} \in \mathbb{Z}_{\geq 0}$. This is given by a *weighted bipartite graph*. The two sets of vertices consist of chemical species S_i , $i = 1, \dots, m$, and complexes C_j ,

$j = 1, \dots, n$. For each edge $\{S_i C_j\}$ there is a weight $y_{ij} \in \mathbb{Z}_{\geq 0}$. This defines the weighted adjacency matrix

$$\begin{pmatrix} 0 & Y_s \\ Y_s^t & 0 \end{pmatrix},$$

with $Y_s = (y_{ij})_{i=1, \dots, m, j=1, \dots, n}$. We denote the columns of Y_s by $y_1, \dots, y_n \in (\mathbb{Z}_{\geq 0})^m$. Then $N = Y_s I_a$ since I_a contains exactly one -1 and one 1 in each column. This means differences of columns of Y_s are formed. But the columns of Y_s contain the coefficients of complexes in (3). Defining $\Psi_j(x) = x^{y_j}$, $j = 1, \dots, n$, we have $\vec{v}(x) = I_K \Psi(x)$.

For a generalized mass action system the exponents differ from y_{ij} in some places. This can be incorporated in this model by choosing a second set of weights η_{ij} on the edges of the bipartite graph. This defines a second weighted adjacency matrix with block $Y_k \in \mathbb{Z}^{m,n}$. We denote the columns of Y_k by $\eta_1, \dots, \eta_n \in \mathbb{Z}^m$ and use $\Psi_j(x) = x^{\eta_j}$, $j = 1, \dots, n$.

Altogether, this leads to the following Laurent polynomial equations

$$0 = Y_s I_a I_K \Psi(x), \quad \text{with } \Psi(x) = \begin{pmatrix} x^{\eta_1} \\ \vdots \\ x^{\eta_n} \end{pmatrix}. \quad (5)$$

We are interested in positive (or at least non-negative) solutions.

Remark 2.2. In this application, we can state some obvious restrictions on the graphs. For the weighted directed graph we assume that for each vertex C_i there is at least one incident edge. An oriented edge $C_j \rightarrow C_i$ and its opposite $C_i \rightarrow C_j$ with two associated constants k_{ij} and k_{ji} are simultaneously possible. They are called reversible reactions. But there are no parallel edges. For the bipartite graph, we assume that Y_s and Y_k have the same support (i.e., $y_{ij} = 0$ iff $\eta_{ij} = 0$) and that $\eta_{ij} \leq y_{ij}$, $\forall i = 1, \dots, m, j = 1, \dots, n$. Moreover, we assume that $y_i \neq y_j$ or $\eta_i \neq \eta_j$ for all $i \neq j$. Additionally, we assume that every vertex S_i is adjacent to at least one vertex C_j . But there may be C_j without incident edges.

System (5) describes the steady state solutions of a dynamical system. This has flow-invariant subspaces of the form $(x_0 + \text{im}(Y_s I_a I_K)) \cap (\mathbb{R}_{\geq 0})^m$, see [26] or Lemma 2.1 in [21]. Thus linear restrictions

$$v_i^t x - c_i = 0, \quad (6)$$

$i = 1, \dots, m - \text{rank}(Y_s I_a I_K)$, are satisfied in addition. Here, the v_i form an orthonormal basis of the orthogonal complement of $\text{im}(Y_s I_a I_K)$ and the $c_i \in \mathbb{R}$ are given constants. Physically, the meaning is conservation of mass, or moiety, or other quantities.

Example 2.1 (continued). The weighted directed graph and weighted bipartite graph of this example are presented in Fig. 2. There are $n = 11$ complexes and

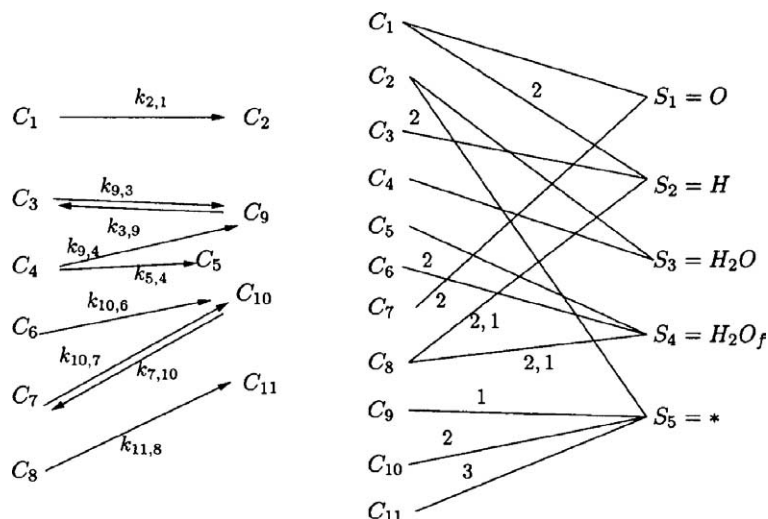


Fig. 2. Weighted directed graph and weighted bipartite graph, describing the chemical reaction from Example 2.1. An edge in the bipartite graph without number has weights $y_{ij} = \eta_{ij} = 1$. If only one number is associated to the edge, then $y_{ij} = \eta_{ij}$, else $y_{ij} > \eta_{ij}$.

$$Y_s = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 0 & 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 3 \end{bmatrix},$$

$$\Psi(x) = (x_1 x_2^2, x_3 x_5, x_2^2, x_3, x_4, x_4^2, x_1^2, x_2 x_4, x_5, x_5^2, x_5^3)^t.$$

The equations for the steady states are

$$\begin{aligned} 0 &= -k_{2,1} x_1 x_2^2 - 2k_{10,7} x_1^2 + k_{7,10} x_5^2, \\ 0 &= -2k_{2,1} x_1 x_2^2 - 2k_{9,3} x_2^2 - 2k_{11,8} x_2 x_4 + 2k_{3,9} x_5, \\ 0 &= k_{2,1} x_1 x_2^2 + (-k_{5,4} - k_{9,4}) x_3, \\ 0 &= k_{5,4} x_3 - 2k_{10,6} x_4^2 - 2k_{11,8} x_2 x_4, \\ 0 &= k_{2,1} x_1 x_2^2 + k_{9,3} x_2^2 + k_{9,4} x_3 + 2k_{10,6} x_4^2 + 2k_{10,7} x_1^2 + 3k_{11,8} x_2 x_4 \\ &\quad - k_{3,9} x_5 - 2k_{7,10} x_5^2, \end{aligned}$$

where the constants k_{ij} are positive real numbers which serve as parameters.

We have $\text{rank}(Y_s I_a I_K) = \text{rank}(Y_s I_a) = 4 < 5 = m$. The additional linear restriction is given by

$$v_1^t x - c_1 = (2 \quad 1 \quad 2 \quad 2 \quad 2) x - c_1 = 0.$$

Remark 2.3. (i) There are equivalent ways of presenting the discrete information of the two graphs. In the chemical literature [6] usually a diagram is drawn. In [36] another graph is introduced. (ii) Also the equations for the steady states can be given in a different form than (4) or (5). For example they can be seen as linear combinations of binomials and monomials.

In the following we will need some matrices Y_r, Y_L, Y_e , whose columns consists of some columns η_j of Y_k . Some monomials x^{η_j} which correspond to pure product complexes C_j do not appear in Eqs. (5), since the corresponding column of I_K is zero. We collect the columns η_j whose complexes C_j appear as reactant complexes into the matrix Y_r . We assume that we have chosen the numbering such that $Y_r = (\eta_1, \dots, \eta_r)$. Some complexes appear as reactant complexes several times. So we list these columns of Y_r several times and denote the matrix by Y_L . Then Y_L has as columns exactly the exponents of $\vec{v}$ in (4). If we add to Y_L the columns η_j of pure product complexes, we denote the new matrix by Y_e .

The graph-theoretical interpretation of Y_r is that the bipartite graph is reduced by removing vertices C_j being pure product complexes and their incident edges. For Y_L this bipartite graph is modified by copying the complexes C_j which appear as reactant complexes and their incident edges according to their multiplicity.

2.2. The approach

In [20] the intersection of a deformed toric variety with a convex polyhedral cone has been introduced as a modification of a general principle from toric geometry.

The convex polyhedral cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$ has been introduced and studied by B. Clarke [6]. Each ray $[z_1 : \dots : z_l]$, $z > 0$, in the cone corresponds to some positive solution of (5) for some value of the parameters k_{ij} . Such a positive solution x is found as a solution of $z = x_0 I_K \Psi(x)$, where we introduced x_0 to compensate the ambiguous length of z . Thus the interior of the convex polyhedral cone corresponds to all positive solutions of system (5) for any value of the parameters k_{ij} .

For fixed values of k_{ij} and given $z > 0$ the system $z = x_0 I_K \Psi(x)$ does not have a solution in general. Conditions on z need to be satisfied.

Definition 2.4 [20]. The ideal

$$I_{\hat{Y}_L}^{\text{def}} = \{f \in Q(\underline{k})[z] \mid f(x_0 I_K \Psi(x)) = 0, \forall x_0 \in \mathbb{C}, x \in (\mathbb{C} \setminus \{0\})^m\}$$

is called *deformed homogeneous toric ideal*.

For each choice of positive real k_{ij} , the deformed toric ideal reduces to a homogeneous ideal in $\mathbb{R}[z]$. In this special case, we call the associated variety $V(I_{\hat{Y}_L}^{\text{def}}) \subseteq \mathbb{P}^{l-1}$ *deformed projective toric variety*. Different choices of values for k_{ij} give a parameterization of the family of projective varieties.

The names *deformed homogeneous toric ideal* and *deformed projective toric variety* are justified by the close relation to the homogeneous toric ideal $I_{\hat{Y}_k} = \{f \in Q[w] \mid f(x_0 x^{\eta_1}, \dots, x_0 x^{\eta_n}) = 0, \forall x, x_0\}$. The torus action of the algebraic torus $T = (\mathbb{C} \setminus \{0\})^m$

on $\mathbb{C}^m$ by $x \mapsto (t_1 x_1, \dots, t_m x_m)$ induces a torus action on $\mathbb{C}^n$ by the matrix Y_k . For fixed $\underline{k}$, the evaluation at $e = (1, \dots, 1)$ gives $I_K \Psi(e)$. The point generates an orbit $\mathcal{O}_{I_K \Psi(\underline{1})}$ under the torus action. $V(I_{\hat{Y}_L}^{\text{def}})$ is the Zariski closure of this orbit in the projective space. Thus $I_{\hat{Y}_L}^{\text{def}}$ is a prime ideal in $\mathcal{Q}(\underline{k})[z]$, which also follows from Corollary 2.6 in [13].

The family of deformed projective toric varieties covers the cone

$$\{[z_1 : \dots : z_l] \in \mathbb{P}^{l-1} \mid z > 0, z \in \ker(Y_s I_a)\}.$$

That means for each ray $[z]$ in the interior of the cone there are values of k_{ij} such that $[z] \in V(I_{\hat{Y}_L}^{\text{def}})$. For $[z] \in (I_{\hat{Y}_L}^{\text{def}})$ the system $z = x_0 I_K \Psi(x)$ has at least one positive solution $x \in (\mathbb{R}_+)^m$.

The efficient computation of a Gröbner basis of $I_{\hat{Y}_L}^{\text{def}}$ is done in the following way. Since a vertex C_j may be the initial vertex for several arrows (reactions), there will be linear equations in $I_{\hat{Y}_L}^{\text{def}}$. For example, if the μ th arrow is $C_j \rightarrow C_i$, and the ν th arrow is $C_j \rightarrow C_s$, then $k_{ij} z_\nu - k_{sj} z_\mu \in I_{\hat{Y}_L}^{\text{def}}$. Remember from Section 2.1 that $Y_r = (\eta_1, \dots, \eta_r)$ denotes the matrix of reactant complexes. A Gröbner basis of $I_{\hat{Y}_r} = \{f \in \mathcal{Q}[w] \mid f(x_0 x^{\eta_1}, \dots, x_0 x^{\eta_r}) = 0, \forall x_0, x\}$ is computed. Implementations of special efficient algorithms are available for example in “Singular” [23]. Some of these algorithms are described in [34, Chapter 12]. In particular, $w^{g^+} - w^{g^-} \in I_{\hat{Y}_r}$ corresponds to $g = g^+ - g^- \in \ker(\hat{Y}_r)$. The columns of Y_L are the columns of Y_r , but some are repeated. Thus reordering of g and filling up with zeros gives $\bar{g} \in \ker(\hat{Y}_L)$. This way, the Gröbner basis of $I_{\hat{Y}_r}$ is converted into polynomials in $I_{\hat{Y}_L}^{\text{def}}$ using $z = I_K w$. Together with the linear polynomials, this gives the desired Gröbner basis of $I_{\hat{Y}_L}^{\text{def}}$.

The cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$ has been investigated a lot by chemists and they implemented algorithms for the computation of a set of minimal generators (see, for example, [29] or [26, Chapter 3] and references therein). Also algorithms from computational geometry [1] are applicable. Intersecting a convex polyhedral cone with a generic hyperplane defines a polytope, whose vertices are enumerated in [1] using the Simplex algorithm and an associated tree. In contrast, the structure of the matrix $Y_s I_a$ is exploited in [29]. The basic idea is that this cone is the smallest cone in a nested chain of cones.

As pointed out in [20], there are two different types of minimal generators of the convex polyhedral cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$. Since $\ker(I_a) \subseteq \ker(Y_s I_a)$, we have also for the cones

$$\ker(I_a) \cap (\mathbb{R}_{\geq 0})^l \subseteq \ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l.$$

Since I_a is an incidence matrix of a directed graph, minimal generators of $\ker(I_a) \cap (\mathbb{R}_{\geq 0})^l$ are *positive circuits* (oriented cycles) of the directed graph. In many cases these positive circuits are also minimal generators of the cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$. For chemical reaction systems, in many cases positive circuits occur as pairs of forward–backward reactions.

Definition 2.5. A minimal generator of the convex polyhedral cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$, which is not a positive circuit of the directed graph, is called *stoichiometric generator*.

The positive circuits are given by the structure of the directed graph. But also the existence of stoichiometric generators is influenced by the directed graph. Besides oriented cycles of the directed graph, we also need the notion of *connected component*, which is a minimal set of vertices which are connected by paths, and *strong connected component*, which is a minimal set of vertices such that each pair of vertices lies on an oriented cycle. A *terminal strong connected component* is a strong connected component with the property that no arrow is pointing out.

These notions are fundamental in the work by Feinberg and others [15,24,30]. They investigate the matrix $I_a I_K$ which is similar to a *Laplacian matrix* (see [4, p. 27]) and thus has similar properties as shown in [18], see also [21].

By a support of a vector one usually means the components being unequal zero. For a vector in the kernel of $Y_s I_a$, we need a modified definition. Given a vector $v \in \mathbb{C}^l$ by

$$S_v = \{j \in \{1, \dots, n\} \mid \exists \mu \in \{1, \dots, l\} \text{ with } (I_K e_j)_\mu \neq 0 \text{ and } v_\mu \neq 0\},$$

we define the set of indices of initial vertices of arrows (reactant complexes) in the directed graph, to form the support of v .

Lemma 2.6. *Assume there exists a stoichiometric generator E_i such that the support S_{E_i} is contained in strong connected components. For each connected component, let one of the two conditions is satisfied. Either S_v contains at most one entry in this connected component, or the positive circuits are forward–backward reactions. Then there is another stoichiometric generator which depends linearly on E_i modulo positive circuits of the directed graph.*

Proof. For each non-zero entry in $-E_i$, there is a positive circuit having a positive entry at this position. Thus, a linear combination results in a new stoichiometric generator. $\square$

Please observe that the first assumptions are satisfied, if the directed graph has only connected components which are strongly connected and $\text{rank}(Y_s I_a) < \text{rank}(I_a)$.

The aim of this paper is to clarify the role of these two types of generators. The stoichiometric generators are investigated in Section 4. In Section 3, we relate the positive circuits to the existence of solutions in the limit with zero-components or at infinity. These are projective solutions of certain subsystems of the polynomial system (5).

Definition 2.7. For a subgraph of the directed graph, we define a *subgraph subsystem* by deleting the columns of I_a , which do not correspond to arrows in the subgraph, and deleting rows in I_K giving a reduced system $Y_s \tilde{I}_a \tilde{I}_K \Psi(x) = 0$. If the subgraph is a positive circuit, then we call it *circuit subsystem*. If the subgraph is given by a stoichiometric generator, then we call it *stoichiometric subgraph system*.

The next section deals with circuit subsystems.

3. Real solutions at infinity

In this section, we derive results for the chemical reaction system (5) which are analog to Bernstein's second theorem for general systems of sparse Laurent polynomials (see [2] and Theorem 6.1 in the appendix of [25]). We start by recalling the theory.

3.1. Bernstein's second theorem

We consider the sparse polynomials

$$f_i(x) = \sum_{j=1}^n c_{ij} x^{a_j}, \quad i = 1, \dots, r, \quad (7)$$

where $c_{ij} \in \mathbb{C} \setminus \{0\}$ are given constants. The exponents $a_1, \dots, a_n \in \mathbb{Z}^m$ of the monomials $x^{a_1}, \dots, x^{a_n} \in \mathbb{C}[x_1, \dots, x_m]$ are given fixed. These exponents form the so-called *support* of the sparse polynomials. The convex hull $\text{conv}(a_1, \dots, a_n)$ of the exponents is called *Newton polytope*. We assume in the following that the Newton polytope has dimension m which is the number of variables. (If this is not the case, the Hermite normal form may be used to compute a basis of the lattice and reduce to this situation.) A *facet* of the Newton polytope is a face of dimension $m - 1$. A facet F is characterized by a linear functional $\omega: \mathbb{R}^m \rightarrow \mathbb{R}$, $x \mapsto \omega^t x$ (where $\omega \in \mathbb{R}^m$ is a given vector) such that the linear functional ω attains its minimum at the facet. This *inner normal* is unique up to scaling. By an *initial facet system* we mean

$$\text{in}_\omega(f_i) = \sum_{a_j \in F_\omega} c_{ij} x^{a_j} = 0, \quad i = 1, \dots, r,$$

which is derived from the original polynomials f_i , $i = 1, \dots, r$, by neglecting the monomials whose exponents are not on the facet F_ω of the Newton polytope. Similarly, an *initial face system* is defined.

Analogue definitions hold if the supports of the polynomials f_i are different. Assume that there are supports $\mathcal{A}_i$, $i = 1, \dots, r$, and thus r Newton polytopes. Assume there are k_i polynomials with the same support:

$$f_{ij}(x) = \sum_{a \in \mathcal{A}_i} c_{ja} x^a, \quad i = 1, \dots, r, \quad j = 1, \dots, k_i.$$

A *face* of the tuple of polytopes $\text{conv}(\mathcal{A}_i)$, $i = 1, \dots, r$, is a set of subsets $F = (F_1, \dots, F_r)$ with $F_i \subseteq \text{conv}(\mathcal{A}_i)$, $i = 1, \dots, r$, such that a linear functional ω attains for each $i = 1, \dots, r$ its minimum over $\text{conv}(\mathcal{A}_i)$ at F_i . If the dimension of the face differs from the

dimension of the tuple of polytopes by one, then the face is called *facet*. Without loss of generality we assume that the dimension of the Minkowski sum

$$\sum_{i=1}^r \text{conv}(\mathcal{A}_i) = \left\{ \sum_{i=1}^r b_i \mid b_i \in \text{conv}(\mathcal{A}_i) \right\}$$

is m . By an *initial facet system* we mean

$$\text{in}_\omega(f_{ij}) = \sum_{a \in F_i} c_{ja} x^a = 0, \quad j = 1, \dots, k_i, \quad i = 1, \dots, r.$$

Analogously, *initial face systems* are defined.

The *mixed volume* $MV(\text{conv}(\mathcal{A}_1), \dots, \text{conv}(\mathcal{A}_r), k_1, \dots, k_r)$ is defined as a coefficient of a certain polynomial [7, p. 322]. By Bernstein's theorem [7, p. 331] it is generically the number of complex solutions in $(\mathbb{C} \setminus \{0\})^m$ of a system $f_{ij}(x) = 0$, $i = 1, \dots, r$, $j = 1, \dots, k_i$ with $k_1 + \dots + k_r = m$.

In the following theorem we make the restriction *generically* more precise.

Theorem 3.1 ([2], Theorem 6.1 in [25]). *Given a system of sparse Laurent polynomials $f_1, \dots, f_m$ in m variables with support $\mathcal{A}_1, \dots, \mathcal{A}_r$ where the supports arise with multiplicities $k_1, \dots, k_r$. The number of toric solutions $x \in (\mathbb{C} \setminus \{0\})^m$ (counted with multiplicity) equals the mixed volume if and only if the initial facet systems have no non-trivial solutions in $\mathbb{C}^m$ or at infinity.*

Remark 3.2. (a) In [25] the existence of non-trivial solutions is formulated more precisely in terms of a vanishing sparse resultant. (b) In the original work [2] Bernstein formulated this criterion for faces instead of facets and demanded the non-existence of non-trivial affine solutions. (c) Clarke [5] calls the Newton polytope exponent polytope and derives similar results. These results are also recalled in [6, Chapter VI] but are applied in [6] in a different way than we do in this article.

Sketch of Proof. First we sketch the constructive proof for the case of one support $\mathcal{A}$ forming the columns of a matrix $\mathcal{A}$. For simplicity we assume that the dimension of the Newton polytope (and thus also the lattice) is m . The solutions $x \in (\mathbb{C} \setminus \{0\})^m$ may be found from a line in the intersection of the kernel of the coefficient matrix with the homogeneous toric variety $V(I_{\hat{\mathcal{A}}})$. Here $\hat{\mathcal{A}}$ is the matrix whose columns are $(1, a_i)$, with a_i the exponents of monomials x^{a_i} and

$$I_{\hat{\mathcal{A}}} = \{ f \in \mathbb{C}[z_1, \dots, z_n] \mid f(x_0 x^{a_1}, \dots, x_0 x^{a_n}) = 0, \forall x \in (\mathbb{C} \setminus \{0\})^m, x_0 \in \mathbb{C} \},$$

the homogeneous toric ideal. Rescalings $\tau \in (\mathbb{C} \setminus \{0\})^m$, $(x_0, x_1, \dots, x_m) \mapsto (x_0, \tau_1 x_1, \dots, \tau_m x_m)$ defines a torus action which induces by the given monomials a torus action on $\mathbb{C}^n$. $V(I_{\hat{\mathcal{A}}}) \subseteq \mathbb{P}^{n-1}$ is the Zariski closure of the maximal orbit $\mathcal{O}_1$ of this induced torus action. (The quotient ring $\mathbb{C}[z]/I_{\hat{\mathcal{A}}}$ is not assumed to be normal as usually in algebraic geometry. This property is not used in this article.)

If z is generating an intersecting line such that all components z_i are unequal zero then at least one solution $x \in (\mathbb{C} \setminus \{0\})^m$ of (7) is found by solving $x_0 x^A = z$ by Hermite normal form or Smith normal form. If z is real and positive then one associated x is positive.

The cases with some $z_i = 0$ are related to the facets of the Newton polytope as follows. Assume $\text{conv}(\mathcal{A})$ has N facets with inner normals $\omega_1, \dots, \omega_N$. Each inner normal $\omega_i \in \mathbb{Z}^m$ is normalized such that the components $\omega_{i1}, \dots, \omega_{im}$ are relatively prime. Denote by V the matrix with rows ω_i . In [7, p. 309] the facet variables $t_1, \dots, t_N$ are introduced by

$$\begin{aligned} x_1 &= t_1^{\omega_{11}} \cdot \dots \cdot t_N^{\omega_{N1}}, \\ &\vdots \\ x_m &= t_1^{\omega_{1m}} \cdot \dots \cdot t_N^{\omega_{Nm}}. \end{aligned} \quad (8)$$

After substitution into the monomials, we obtain new monomials $t^{V_{a_1}}, \dots, t^{V_{a_n}}$ in the facet variables, whose exponents form the columns of VA . Then $I_{\widehat{VA}} = I_{\hat{A}}$. Since a shift of the Newton polytope does not change the homogeneous toric variety one may multiply all monomials by t^b where $-b_j$ is the minimal value of the functional $\mathbb{R}^m \rightarrow \mathbb{R}, w \mapsto v_j^t w$ on $\text{conv}(\mathcal{A})$. Then $V_{a_i} + b \geq 0$ and $(V_{a_i} + b)_j = 0$ iff a_j is on the j th facet. That means that $(\mathbb{C} \setminus \{0\})^m \rightarrow \mathbb{P}^{n-1}, (x_1, \dots, x_m) \mapsto [x^{a_1} : \dots : x^{a_n}]$ and $(\mathbb{C} \setminus \{0\})^N \rightarrow \mathbb{P}^{n-1}, (t_1, \dots, t_N) \mapsto [t^{V_{a_1}+b} : \dots : t^{V_{a_n}+b}]$ are two different parameterizations of the maximal orbit $\mathcal{O}_1 \subset V(I_{\hat{A}})$.

The torus group $(\mathbb{C} \setminus \{0\})^N$ is operating non-faithful on $\mathbb{P}^{n-1}$ since the maximal orbit has dimension m . Representatives of $(\mathbb{C} \setminus \{0\})^N / (\mathbb{C} \setminus \{0\})^m$ are chosen by setting some $t_i = 1$ in the following way. For a vertex a_j of the Newton polytope we distinguish neighboring facets and non-neighboring facets. All facet variables of non-neighboring facets are chosen to be one. If the inner normals of the neighboring facets are linearly dependent this dependence defines algebraic relations in the facet variables. By this choice $t^{V_{a_j}+b} = 1$. Substitution of $z_j = 1$ into $I_{\hat{A}}$ gives an affine toric ideal.

In [8] it is explained that not too many t_i may vanish at the same time since $[0 : \dots : 0]$ is not an element of $\mathbb{P}^{n-1}$. The monomial ideal $B = \langle \{\prod_{i \notin \sigma(1)} t_i \mid \sigma \text{ is a face of the Newton polytope}\} \rangle$ is generated by monomials where $\sigma(1)$ is the set of facets which have σ as a face. Then

$$\mathbb{C}^N \setminus V(B) \rightarrow \mathbb{P}^{n-1}, \quad (t_1, \dots, t_N) \mapsto [t^{V_{a_1}+b} : \dots : t^{V_{a_n}+b}]$$

is a parameterization of $V(I_{\hat{A}})$.

For $t_j = 0$ we have $t^{V_{a_i}+b} = 0$ for a_i , not on the j th facet. This reduces the original system $f(x) = 0$ to an initial facet system. This gives a homogeneous toric subvariety corresponding to $a_{v_1}, \dots, a_{v_s}$, the points on the j th facet. Assume there exists $z \in \mathbb{C}^n$ in the kernel of the coefficient matrix with $z_\mu = 0$ for all $\mu \notin \{v_1, \dots, v_s\}$ and $[z] \in V(I_{\hat{A}})$. Then Hermite normal form gives solutions $t \in \mathbb{C}^N$ of

$$t_0 t^{V_{a_1}+b} = z_1, \quad \dots, \quad t_0 t^{V_{a_n}+b} = z_n.$$

For backsubstitution with (8) we eventually need projective coordinates $[x_0 : x_1 : \dots : x_m] \in \mathbb{P}^m$. For an index j of a facet with $t_j = 0$ and $d_j = \min_{\mu \in \{1, \dots, m\}} \omega_{j\mu} < 0$ rescaling in projective space $\mathbb{P}^m$ gives

$$\begin{aligned} x_0 &= t_j^{-d_j}, \\ x_1 &= t_1^{\omega_{11}} \cdot \dots \cdot t_N^{\omega_{N1}} t_j^{-d_j}, \\ &\vdots \\ x_m &= t_1^{\omega_{1m}} \cdot \dots \cdot t_N^{\omega_{Nm}} t_j^{-d_j}. \end{aligned}$$

Substitution of $t_j = 0$ gives a (generalized) solution with some components being zero or at infinity ($x_0 = 0$). If the solution t is real then $[x] \in P^m$ is real. A real non-negative solution t with some zero-components corresponds to a real non-negative solution $[x] \in \mathbb{P}^m$ with zero components or at real infinity with non-negative real components.

This corresponds to a solution of the initial facet system $\text{in}_{\omega_j}(f_i) = 0$, $i = 1, \dots, r$. Since the initial facet system is weighted homogeneous solutions may also be interpreted in the weighted projective space $\mathbb{P}(\omega_{j1}, \dots, \omega_{jm})$ where points are identified under the one-dimensional torus action

$$x \mapsto (t_j^{\omega_{j1}} x_1, \dots, t_j^{\omega_{jm}} x_m).$$

Let us investigate closer the case, where $f(x) = 0$ has a curve of non-negative real solutions. By homogeneity, the initial facet system has a path of solutions $p(t_j) = (t_j^{\omega_{j1}} a_1, \dots, t_j^{\omega_{jm}} a_m)$, $t_j \in (0, \infty)$, with $a \in (\mathbb{R}_{\geq 0})^m \setminus \{0\}$ a fixed solution. If the solutions of the original system form a path as well then a local parameterization $x(t_j)$, $t_j \in (0, \varepsilon)$, satisfies

$$\lim_{t_j \rightarrow 0} \left(\frac{x_1(t_j)}{p_1(t_j)}, \dots, \frac{x_m(t_j)}{p_m(t_j)} \right) = (1, \dots, 1).$$

In the original work [2] the function $x(t_j)$ is a Puiseux series. Its exponents of the initial terms are given by the inner normal ω .

We briefly discuss the case that several facet variables vanish, i.e., $t_{j_1} = \dots = t_{j_s} = 0$. Then the initial face system is multihomogeneous with respect to $\omega_{j_1}, \dots, \omega_{j_s}$ and the solutions are orbits of the s -dimensional torus action, i.e., are elements of $(\mathbb{C} \setminus \{0\})^m / (\mathbb{C} \setminus \{0\})^s$. But the limit $\lim_{t_{j_1} \rightarrow 0, \dots, t_{j_s} \rightarrow 0} [1 : x(t)]$ does not exist in $\mathbb{P}^m$ in general. The theory in [8] explains that this happens if the inner normal fan is not simplicial. Then we use the following trick. The problem is embedded into a larger problem. Additionally, each equation is multiplied with each x_i . Of course this enlarged polynomial system has the same solutions than the original system. The Newton polytope of the new system is the Minkowski sum of the old Newton polytope and the m -simplex. The projective toric variety associated to the Minkowski sum contains both $V(I_{\hat{A}})$ and $\mathbb{P}^m$ as embeddings. Each face of the old Newton polytope gives rise to a face of the new Newton polytope. The same holds true for the m -simplex. Thus we have

$$\begin{aligned} x_1 &= t_1^{\omega_{11}} \cdot \dots \cdot t_N^{\omega_{N1}} \cdot \tilde{x}_1 / \tilde{x}_0, \\ &\vdots \\ x_m &= t_1^{\omega_{1m}} \cdot \dots \cdot t_N^{\omega_{Nm}} \cdot \tilde{x}_m / \tilde{x}_0, \end{aligned}$$

where $\tilde{x}_0, \tilde{x}_1, \dots, \tilde{x}_m$ are the facet variables of the m -simplex. Consider again the situation that an initial face system of the old Newton polytope has a non-trivial solution. This face gives rise to a face of the new Newton polytope, but with different neighboring facets. Some neighboring facets arise from the m -simplex, giving conditions $\tilde{x}_i = 0$ for some $i \in \{0, 1, \dots, m\}$. If the limits are uniquely defined (equivalently the inner normal cones are simplicial) then Puiseux series exist.

For several supports $\mathcal{A}_1, \dots, \mathcal{A}_r$, this technique has to be generalized. Finding solutions $x \in \mathbb{C}^m$ of $f_i(x) = 0, i = 1, \dots, r$, is equivalent to finding solutions $x \in \mathbb{C}^{m+r-1}$ of $f_1(x) + x_{-2}f_2(x) + \dots + x_{-r}f_r(x) = 0$ where $x_{-2}, \dots, x_{-r}$ are arbitrary. The multihomogeneous toric ideal

$$I_{\check{A}} = \{f \in \mathbb{C}[z] \mid f(x_0 x^{a_1}, \dots, x_0 x_{-i} x^{a_j}, \dots, x_0 x_{-r} x^{a_n}) = 0, \forall x, x_0, x_{-i}\},$$

and its variety are used. The lifting is

$$\check{A} = \begin{pmatrix} 1 & \dots & 1 & & & & \dots & 1 & \dots & 1 \\ 0 & \dots & 0 & 1 & \dots & 1 & & 0 & \dots & 0 \\ \vdots & & & & & & \ddots & & & \vdots \\ 0 & \dots & & & & & \dots & 1 & \dots & 1 \\ a_1 & \dots & & & & & & \dots & \dots & a_n \end{pmatrix}.$$

The facet variables are now defined by the facets of the tuple of Newton polytopes.

Remark 3.3. (a) The proof nicely reflects the structure of the toric variety as a Zariski closure of a maximal orbit. The boundary consists of toric subvarieties which are Zariski closures of orbits of lower dimension. (b) The proof also shows that the protective toric variety is glued together from affine toric varieties. For each vertex of the Newton polytope there is an affine toric variety. All these are identified on the maximal orbit. (c) The one-dimensional torus action $x \mapsto (t_j^{\omega_{j1}} x_1, \dots, t_m^{\omega_{jm}} x_m)$ has been used numerically in [27]. (d) The technique for several Newton polytopes is usually referred to as Cayley trick [22].

3.2. Applications with one Newton polytope

For the system of equations (5), we know about real non-negative solutions of subgraph subsystems by the knowledge of the minimal generators of the convex polyhedral cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$. If such a circuit subsystem is an initial face system, then we found a real solution with some components zero or at infinity (in the real projective space).

Recall that the columns of Y_k are called $\eta_1, \dots, \eta_n$. The Newton polytope of $Y_s I_a \Psi(x) = 0$ is the convex hull of $\{\eta_i \mid \exists C_i \rightarrow C_j\}$. In other words the Newton polytope is the convex hull of the columns of Y_r .

Definition 3.4. Take a vector $v \in \mathbb{C}^l$ with some components being zero and recall the definition $S_v = \{j \in \{1, \dots, n\} \mid \exists \mu \in \{1, \dots, l\} \text{ with } (I_K e_j)_\mu \neq 0 \text{ and } v_\mu \neq 0\}$ of initial vertices associated to the support of v . For $v = I_K \Psi(x)$ these vertices correspond to monomials. We call x^{η_j} , $j \in S_v$, the *support monomials* of v .

For minimal generators E_i of the convex polyhedral cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$, we have obviously $\{\eta_j \mid j \in S_{E_i}\} \subset \{\eta_i \mid \exists C_i \rightarrow C_j\}$, i.e., all support monomials appear in $Y_s I_a I_K \Psi(x) = 0$.

Theorem 3.5. Consider system (5) of sparse polynomials $Y_s I_a I_K \Psi(x) = 0$ given by a weighted directed graph and a weighted bipartite graph as defined in Section 2. Let $\text{conv}(\{\eta_i \mid \exists C_i \rightarrow C_j\}) = \text{conv}(\eta_1, \dots, \eta_r)$ be the Newton polytope. Let E_i be a minimal generator of the convex polyhedral cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$ and assume the following conditions.

- (i) E_i is a positive circuit in the directed graph.
- (ii) There is a face F of the Newton polytope such that $F \cap \{\eta_1, \dots, \eta_r\} = -\{\eta_j \mid j \in S_{E_i}\}$. That means the support of E_i forms a face.
- (iii) The points in the support $\{\eta_j \mid j \in S_{E_i}\}$ are affine linearly independent.
- (iv) Each vertex of the directed graph associated to a support monomial of E_i is the initial vertex of an arrow exactly once. That means the support of E_i forms a terminal strong connected component of the directed graph.

Then the system $Y_s I_a I_K \Psi(x) = 0$ has a non-negative real solution with some components zero or infinity.

Proof. The proof is a variation of the proof of Theorem 3.1.

The nullspace $\ker(Y_s I_a)$ intersects the deformed toric variety on coordinate hyperplanes. By (iii) the projective toric variety for this face is just the projective space itself. By (iv) the linear equations in the deformed toric ideal vanish for the face. By (i) there is a line in the intersection of the homogeneous deformed toric variety with the kernel of the coefficient matrix. Since the generator E_i of the line is non-negative the solutions of the facet variables are non-negative and thus the x_i are non-negative. $\square$

Next we give criteria for condition (ii), i.e., for the support S_{E_i} to be a face. In general, this is a difficult combinatorial task. But here we exploit the additional structure of sparsity of monomials. Analogous to the support of a vector, we define the notion of the support of a monomial.

Definition 3.6. For a Laurent monomial x^a in the variables $x_1, \dots, x_m$ the support is $T(x^a) = \{j \in \{1, \dots, m\} \mid a_j \neq 0\}$ and the x_j with $j \in T(x^a)$ are the *support variables*.

Lemma 3.7. Given $\eta_i \in (\mathbb{Z}_{\geq 0})^m$, $i = 1, \dots, r$, and $S \subset \{1, \dots, r\}$ with a set of points η_i , $i \in S$. Suppose the supports $T(x^{\eta_i})$ for $i \in S$ and for $i \in \{1, \dots, r\} \setminus S$ satisfy the following statement.

(C) A support of a monomial outside of S is not contained in the union of supports of S . That means $T(x^{\eta_i}) \not\subseteq \bigcup_{j \in S} T(x^{\eta_j})$, $\forall i \in \{1, \dots, r\} \setminus S$.

Then the points η_j , $j \in S$, generate a face of the Newton polytope $\text{conv}(\eta_1, \dots, \eta_r)$. More precisely, there is a face F of the Newton polytope such that the intersection is $F \cap \{\eta_1, \dots, \eta_r\} = \{\eta_j \mid j \in S\}$.

Proof. We construct explicitly a linear functional ω which attains its minimum at the points of S and is larger outside. For all support variables x_k of monomials x^{η_j} , $j \in S$, we set $\omega_k = 0$. For all other variables we set $\omega_k = 1$. Then $\omega^t \eta_j = 0$ for all $j \in S$. For η_i , $i \notin S$, there is a variable x_k with $\omega_k > 0$ and $(\eta_i)_k > 0$ by condition (C). Thus the support of S forms a face. $\square$

Remark 3.8. (a) An equivalent formulation for (C) is the condition that all points associated to S lie in one coordinate hyperplane and the others above. (b) For chemical reaction systems the set S is a collection of complexes (vertices of the bipartite graph). Condition (C) may also be formulated as a property of the reduced bipartite graph associated to Y_r . (C) means that the set of vertices S_i decomposes into two subset. One set of vertices being adjacent to a vertex C_j with $j \in S$ and one subset of vertices which are not adjacent.

Lemma 3.9. Given $\eta_i \in (\mathbb{Z}_{\geq 0})^m$, $i = 1, \dots, r$, and $S \subset \{1, \dots, r\}$ with a set of points η_i , $i \in S$. Suppose the supports $T(x^{\eta_j})$ for $j \in S$ and for $i \in \{1, \dots, r\} \setminus S$ satisfy the statements (D) and either (C^+) or (C_-) .

(D) The supports of the monomials in S are all different or all monomials of S have the same weighted degree. That means for all $i, j \in S$ either

- $T(x^{\eta_i}) \cap T(x^{\eta_j}) = \emptyset$ and $T(x^{\eta_i}) \neq \emptyset$ or $T(x^{\eta_j}) \neq \emptyset$ or
- for all $k \in T(x^{\eta_i}) \cap T(x^{\eta_j})$ the degrees satisfy $\deg_k(x^{\eta_i}) = \deg_k(x^{\eta_j})$ and $\deg_W(x^{\eta_i}) = \deg_W(x^{\eta_j})$.

Here, the weighted degree is given by the degree of x_k in a monomial, $\deg_k(x^{\eta_i}) = \eta_{ki}$ and W defines some other weighted degree.

(C⁺) All monomials x^{η_i} outside (i.e., $i \notin S$) have a support which is either not contained in the union of supports of S or the monomial has larger weighted degree. That means for all $i \in \{1, \dots, r\} \setminus S$ either $T(x^{\eta_i}) \not\subseteq \bigcup_{j \in S} T(x^{\eta_j})$ or $\deg_W(x^{\eta_i}) > \deg_W(x^{\eta_j})$, $\forall j \in S$.

(C₋) All monomials x^{η_i} outside ($i \notin S$) have a support which is either not contained in the union of supports of S or the monomial has smaller weighted degree. That means for all $i \in \{1, \dots, r\} \setminus S$ either $T(x^{\eta_i}) \not\subseteq \bigcup_{j \in S} T(x^{\eta_j})$ or $\deg_W(x^{\eta_i}) < \deg_W(x^{\eta_j})$, $\forall j \in S$.

Then the points η_j , $j \in S$, generate a face of the Newton polytope. More precisely, there is a face F of the Newton polytope such that $F \cap \{\eta_1, \dots, \eta_r\} = \{\eta_i \mid i \in S\}$.

Proof. We construct explicitly a linear functional ω which attains its minimum at the points associated to S and which is larger outside. By (D) we can define a weighted degree

by $\deg_W(x_k) = \deg_k(x^{\eta_j})$, $k \in T(x^{\eta_j})$, for all variables of $\bigcup_{j \in S} T(x^{\eta_j})$. Let d be the product of the degrees of the monomials x^{η_j} for the set S . In formulas $d := \prod_{j \in S} \deg_W(x^{\eta_j})$. For each variable x_k in the support $T(x^{\eta_j})$ of $j \in S$ we define $\omega_k = \sigma d / \deg_W(x^{\eta_j})$, where $\sigma = 1$ in case (C^+) is valid and $\sigma = -1$ for the case (C_-) . This is well defined because of condition (D). Then $\omega^t \eta_j = \sigma d$ for $j \in S$.

(C^+) All other variables x_j get $\omega_j = d + 1$. By (C^+) first case for a point η_i outside S there is a variable x_k with $k \notin \bigcup_{j \in S} T(x^{\eta_j})$. For this $\omega_k = d + 1$ and $(\eta_i)_k \neq 0$. Thus $\omega^t \eta_i \geq d + 1$. For the second case $\deg_W(x^{\eta_i}) > \deg_W(x^{\eta_j})$ guarantees $\omega^t \eta_i > d$.

(C_-) For a variable x_v which is not in the union of supports of S , but in a support of a monomial outside S which contains a support of a monomial in S we define $\omega_k = 1$. All other variables get weight zero. $\square$

Remark 3.10. (a) The condition (D) can be reformulated by saying that all points associated to S lie in one coordinate hyperplane. They generate a polytope of dimension less than the dimension of the coordinate hyperplane. Conditions (C^+) and (C_-) mean that all other points either lie above the coordinate hyperplane or lie outside the polytope generated by S . All exceptional points inside the coordinate hyperplane lie all above or below the polytope. (b) For chemical reaction systems the set S is a collection of complexes (vertices of the bipartite graph). Then conditions (D), (C^+) and (C_-) can be formulated as properties of the reduced bipartite graph associated to Y_r . (C^+) and (C_-) mean that the set of vertices S_i decomposes into two subsets, one subset of vertices being adjacent to an C_j , $j \in S$, and the set of the remaining vertices S_k . For a C_j , $j \notin S$, there is an adjacent remaining S_k or the weighted sum of incident edges is different. Condition (D) means that for two vertices C_i and C_j either there is no vertex S_k which is adjacent to both C_i and C_j or if S_k is adjacent to both then the two edges have same weight $\eta_{ki} = \eta_{kj}$ and the sum of weights of incident edges is equal for both C_i and C_j .

The results are summarized in the following way. Under weak additional conditions circuit subsystems are initial face systems.

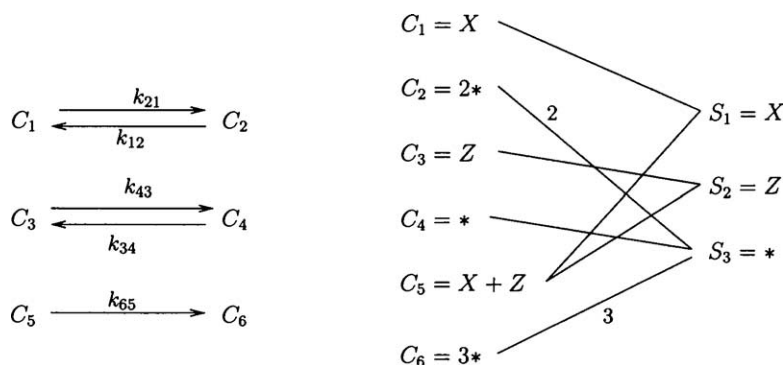


Fig. 3. A small model for reactions on the surface of a metal.

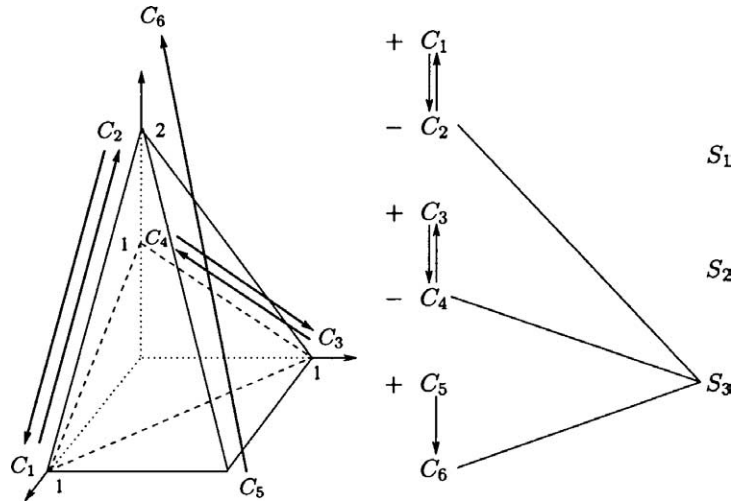


Fig. 4. The Newton polytope and the graph for the signs of the third equation.

Example 3.11. The first example in [10] is illustrated in Fig. 3. The exponents are the columns of the matrix

$$Y_s = Y_k = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 & 0 & 3 \end{pmatrix}$$

giving the equations

$$0 = -k_{2,1}x_1 + k_{1,2}x_3^2 - k_{6,5}x_1x_2,$$

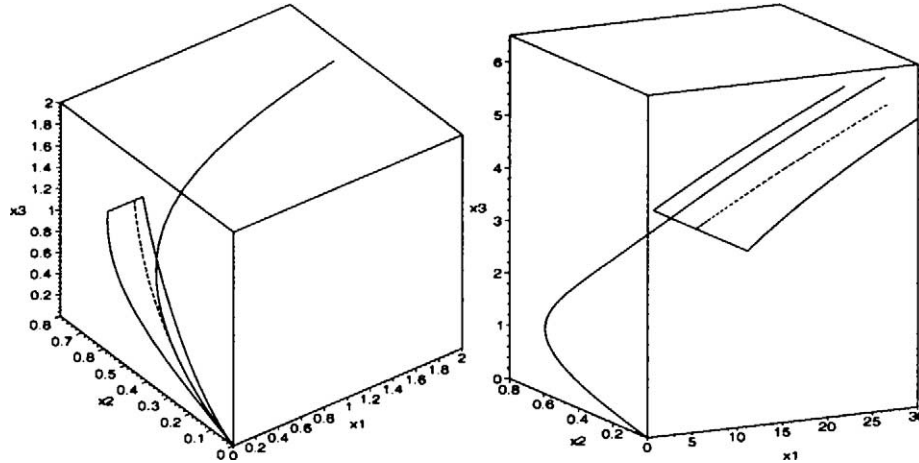
$$0 = -k_{4,3}x_2 + k_{3,4}x_3 - k_{6,5}x_1x_2,$$

$$0 = 2k_{2,1}x_1 - 2k_{1,2}x_3^2 + k_{4,3}x_2 - k_{3,4}x_3 + 3k_{6,5}x_1x_2.$$

The conservation relation is $2x_1 + x_2 + x_3 - c_1 = 0$ with $v_1 = (2, 1, 1)$. The toric ideal is $I_{\hat{Y}_r} = \langle w_1w_2w_3 - w_4^2w_5 \rangle$ while the deformed toric ideal is $I_{\hat{Y}_L}^{\text{def}} = \langle k_{3,4}^2k_{6,5}z_1z_2z_3 - k_{2,1}k_{1,2}k_{4,3}z_4^2z_5 \rangle$. There are two positive circuits, namely $E_1 = (1, 1, 0, 0, 0)$ and $E_2 = (0, 0, 1, 1, 0)$. Each of them gives rise to a face of the Newton polytope $\text{conv}(\eta_1, \dots, \eta_5)$ (Fig. 4). Using facet variables

$$x_1 = t_1t_2t_5^{-2}, \quad x_2 = t_1t_3t_6^{-2}, \quad x_3 = t_1t_4t_5^{-1}t_6^{-1},$$

we compute the irrelevant ideal $B = \langle t_1, t_2t_6, t_3t_5, t_5t_6, t_1t_2t_3 \rangle$ and $V(B) = \{t_1 = t_2 = t_5 = 0 \text{ or } t_1 = t_3 = t_6 = 0 \text{ or } t_1 = t_5 = t_6 = 0\}$. We investigate two limits. For E_1 we choose $t_1 = t_2 = t_6 = 1$ and study the limit $t_3, t_5 \rightarrow 0$ giving the initial face system $k_{2,1}x_1 - k_{1,2}x_3^2 = 0$

Fig. 5. Two limits of a solution curve where all $k_{ij} = 1$.

or equivalently $k_{2,1}t_5^{-2} - k_{1,2}t_5^{-2} = 0$. The solutions form a two-dimensional set since the equations are homogeneous with respect to two weighted gradings. The limit is

$$\lim_{t_3, t_5 \rightarrow 0} \left[1 : \frac{1}{t_5^2} : t_3 : \sqrt{\frac{k_{2,1}}{k_{1,2}}} \frac{1}{t_5} \right] = \lim_{t_3, t_5 \rightarrow 0} \left[t_5^2 : 1 : t_3 t_5^2 : \sqrt{\frac{k_{2,1}}{k_{1,2}}} t_5 \right] = [0 : 1 : 0 : 0].$$

For E_2 we choose $t_3 = t_4 = t_5 = 1$. The limit $t_1, t_2 \rightarrow 0$ gives the initial face system $k_{4,3}x_2 - k_{3,4}x_3 = 0$ or equivalently $k_{4,3}t_6^{-2} - k_{3,4}t_6^{-1} = 0$ yielding $t_6 = k_{4,3}/k_{3,4}$. The limit is the origin.

The chemical interpretation is the following. In the first limit the amount of the first species is infinite such that no space on the surface is left and only the first forward–backward reaction may be performed. In the second limit there is no available space. For very small amount of space first the second reaction pair is performed since $x^{\eta_4} = x_3$ has lower degree than $x^{\eta_2} = x_3^2$.

Example 2.1 (continued). In this example we have

$$Y_k = (\eta_1, \dots, \eta_{11}) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 3 \end{pmatrix}.$$

The Newton polytope is generated by $\eta_1, \eta_3, \eta_4, \eta_6, \eta_7, \eta_8, \eta_9, \eta_{10}$ since C_2, C_5, C_{11} are pure product complexes. The deformed toric ideal $I_{\hat{Y}_L}^{\text{def}}$ is

$$\langle k_{5,4}z_4 - k_{9,4}z_5, k_{11,8}^2z_2z_6 - k_{9,3}k_{10,6}z_9^2, k_{9,3}^2k_{10,7}k_{7,10}z_1^2z_3^2 - k_{2,1}^2k_{3,9}^2z_2^2z_7z_8 \rangle.$$

The convex polyhedral cone is generated by 5 vectors including 3 stoichiometric generators and two positive circuits

$$E_1 = [0, 1, 1, 0, 0, 0, 0, 0], \quad E_2 = [0, 0, 0, 0, 0, 0, 1, 1, 0].$$

Their supports are $S_{E_1} = \{3, 9\}$ and $S_{E_2} = \{7, 10\}$, because $v_2 = k_{9,3}x^{\eta_3}$, $v_3 = k_{3,9}x^{\eta_9}$, $v_7 = k_{10,7}x^{\eta_7}$, $v_8 = k_{7,10}x^{\eta_{10}}$. By Lemma 3.9 the support of E_1 generates a face of the Newton polytope as well as the support of E_2 does. $\omega_1 = (2, 1, 3, 2, 2)$ attains its minimum on the face $\text{conv}(\eta_3, \eta_9)$ while $\omega_2 = (-2, 1, 1, 0, -2)$ attains its minimum on the face $\text{conv}(\eta_7, \eta_{10})$.

The 5-dimensional Newton polytope has 8 facets $F_i, i = 1, \dots, 8$. According to (8), this gives the coordinate transformation with facet variables $t_i, i = 1, \dots, 8$,

$$x_1 = \frac{t_6 t_7}{t_5^2}, \quad x_2 = \frac{t_6 t_8}{t_2 t_5}, \quad x_3 = \frac{t_4 t_6^2}{t_2^2 t_5^4}, \quad x_4 = \frac{t_3 t_6}{t_2 t_5^2}, \quad x_5 = \frac{t_1 t_6^2}{t_2 t_5^2}. \quad (9)$$

In order to determine the exceptional set $V(B)$ we determined the 86 faces and computed a Gröbner basis yielding

$$B = \langle t_3, t_4, t_1 t_6, t_2 t_7, t_5 t_8, t_1 t_2 t_5, t_6 t_7 t_8 \rangle.$$

The monomials in facet variables are

$$\Psi(t) = (t_6 t_7 t_8^2, \frac{t_6^2 t_4 t_1}{t_2 t_5^2}, t_5^2, t_8^2, t_4, \frac{t_2 t_3 t_5^2}{t_6}, t_3^2, t_2^2 t_7^2, t_5 t_8 t_3, t_2 t_5^2 t_1, t_6^2 t_1^2, \frac{t_6^2 t_1^3}{t_2 t_5^2}).$$

The vector v in facet variables is

$$v(t) = (k_{2,1} t_6 t_7 t_8^2, k_{9,3} t_5^2 t_8^2, k_{3,9} t_2 t_5^2 t_1, k_{9,4} t_4, k_{5,4} t_4, k_{10,6} t_3^2, k_{10,7} t_2^2 t_7^2, k_{7,10} t_6^2 t_1^2, k_{11,8} t_5 t_8 t_3).$$

For E_1 we computed that the vertex η_3 lies on the facets $F_1, F_2, F_3, F_4, F_6, F_7$ while η_9 lies on the facets F_3, F_4, F_6, F_7, F_8 . The edge $\{\eta_3, \eta_9\}$ is the intersection of the facets F_3, F_4, F_6, F_7 . We choose a chart of the protective toric variety by restricting to $t_1 = t_2 = t_5 = 1$. Then the initial face system corresponding to E_1 is $k_{3,9}x_2^2 - k_{9,3}x_5 = 0$ or equivalently $k_{9,3}t_8^2 - k_{3,9} = 0$. We substitute $t_3 = t_4 = t_6 = t_7 = 0$ into (9) which yields the limit $x = 0$. The associated constant in the conservation relation is $c_1 = 0$. This agrees with the understanding of chemists. If no free space on the surface is available then the concentrations of all substances on the metal are zero. Since no reactions happen the concentration of H_2O is zero too.

For E_2 the computation yields that η_7 is on the facets $F_1, F_3, F_4, F_5, F_6, F_8$ while η_{10} is on the facets $F_2, F_3, F_4, F_5, F_7, F_8$. The edge $\{\eta_7, \eta_{10}\}$ is the intersection of the facets F_3, F_4, F_5, F_8 . Unfortunately, both inner normal cones of η_7 and η_{10} are not simplicial. Choosing the vertex η_{10} enables us to restrict to $t_1 = t_6 = 1$. The initial face system

$k_{10,7}x_1^2 - k_{7,10}x_5^2$ is equivalent to $k_{10,7}t_7^2 - k_{7,10}t_2^{-2} = 0$ which is equivalent to the equation $v(t) = k_{7,10}E_2$ which implies

$$t_7 = \frac{c}{t_2} \quad \text{with } c = \sqrt{\frac{k_{7,10}}{k_{10,7}}}.$$

This and the linear dependence $-2\omega_2 + \omega_5 + 2\omega_7 - \omega_8 = 0$ of inner normals yields $t_7^2 t_5 = t_2^2 t_8$. The limit to the face is not simply the limit $t_3, t_4, t_5, t_8 \rightarrow 0$, but a limit where $c^2 t_5 = t_2^4 t_8$ is satisfied. We may use the 5-simplex as outlined in the proof of Theorem 3.1. Then the change of coordinates to facet variables is

$$x_1 = \frac{\tilde{x}_1 t_6}{\tilde{x}_0 s_2^2 t_5^2}, \quad x_2 = \frac{\tilde{x}_2 t_6}{\tilde{x}_0 s_1 s_2 t_2 t_5}, \quad x_3 = \frac{\tilde{x}_3 t_6^2}{\tilde{x}_0 s_1 s_2^2 t_2^2 t_5^2}, \quad x_4 = \frac{\tilde{x}_4 t_6}{\tilde{x}_0 s_1 s_2^2 t_2 t_5^2}, \quad x_5 = \frac{\tilde{x}_5 t_6^2}{\tilde{x}_0 s_1 s_2^2 t_2 t_5^2}.$$

The limit to the face involves $\tilde{x}_2, \tilde{x}_3, \tilde{x}_4, s_2 \rightarrow 0$. While the normal cone of one vertex is non-simplicial the other normal cone is. The latter enables us to choose a good coordinate system. We choose a chart with $s_1 = \tilde{x}_1 = \tilde{x}_0 = t_2 = t_5 = t_6 = 1$ leaving $s_2, \tilde{x}_2, \tilde{x}_3, \tilde{x}_4, \tilde{x}_5$. The initial face system in these coordinates is $-k_{10,7} + k_{7,10}\tilde{x}_5^2 = 0$ yielding

$$[x_0 : x_1 : x_2 : x_3 : x_4 : x_5] = [s_2^2 : 1 : s_2 \tilde{x}_2 : \tilde{x}_3 : \tilde{x}_4 : c].$$

Finally, the limit $\tilde{x}_2, \tilde{x}_3, \tilde{x}_4, s_2 \rightarrow 0$ of x is $[0 : 1 : 0 : 0 : 0 : c]$. For the conservation relation this is the limit $c_1 \rightarrow \infty$.

The chemical understanding is that in the limit an infinitely large amount of oxygen O is moving on and from the surface. The concentrations of all other species are relatively small and thus are considered to be zero.

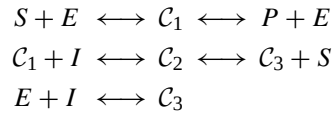
Proposition 3.12. Assume the system (5) of sparse polynomials $Y_s I_a I_K \Psi(x) = 0$ is given by a weighted directed graph and a weighted bipartite graph as defined in Section 2. Let E_i and E_j be two different minimal generators of the convex polyhedral cone $\ker(Y_s I_a) \cap (\mathbb{R}_{\geq 0})^l$ and assume the following conditions.

- (i) E_i and E_j are positive circuits in the directed graph.
- (ii) The exponents of the support-monomials of E_i and E_j generate a face of the Newton polytope.
- (iii) The points in the supports $\{\eta_j \mid j \in S_{E_i} \cup S_{E_j}\}$ are affine linear independent.
- (iv) The supports of E_i and E_j form a terminal strong connected component of the directed graph.

Then the system $Y_s I_a I_K \Psi(x) = 0$ has a non-negative real solution with zero-components or infinity.

More generalizations of Theorem 3.1 including several positive circuits are easily formulated.

Example 3.13. The following chemical reaction is taken from [30] where it is left for further research. It models non-competitive inhibition.



There are 7 species. Thus the Newton polytope is a subset of $\mathbb{R}^7$. But it has dimension 6 and thus the lattice is degenerate. The vector space of conservation vectors has dimension 3. Since the lattice is degenerate the real variety in the positive orthant is always transversal to the affine spaces given by the conservation relation $x_E + x_{C_1} + x_{C_2} + x_{C_3} - c_1 = 0$. The two other relations are

$$x_I + x_{C_2} + x_{C_3} - c_2 = 0, \quad x_S + x_{C_1} + x_P + x_{C_2} - c_3 = 0.$$

The first two components of the directed graph correspond to faces of the Newton polytope by Proposition 3.12 while the last component forms a face by Lemma 3.9. Using the facet variables of the Minkowski sum of the Newton polytope and a 7-simplex we obtain three limits for $[x(t, w_7)] = [x_0 : x_S : x_E : x_{C_1} : x_P : x_I : x_{C_2} : x_{C_3}]$

$$\begin{aligned} \text{1 component:} \quad & \lim_{t_6 \rightarrow 0} \left[w_7 t_6 : \frac{k_{1,2}}{k_{2,1}} w_7 : t_6 \frac{k_{1,2}}{k_{2,1}} : \frac{k_{1,2}}{k_{2,1}} : \frac{k_{3,2}}{k_{2,3}} w_7 : 0 : 0 : 0 \right], \\ \text{2 component:} \quad & x_0 = 0, \quad x_E = 0, \quad x_P = 0, \\ \text{3 component:} \quad & \lim_{t_{14} \rightarrow 0} \left[w_7 t_{14} : 0 : t_{14} \frac{k_{7,8}}{k_{8,7}} : 0 : 0 : w_7 : 0 : 1 \right]. \end{aligned}$$

For the first component the limit corresponds to $c_2 = 0$ while for the last component the limit corresponds to $c_3 = 0$.

We summarize the results of this section as follows. Under weak assumptions, subgraph subsystems corresponding to terminal strong connected components are initial face systems. This means the steady states of the full system (5) for extreme values of the conservation constants c_i , are determined by these subgraph subsystems.

3.3. Several Newton polytopes

In this subsection we distinguish the different supports $\mathcal{A}_i$, $i = 1, \dots, m$, of the single equations in $Y_s I_a I_K \Psi(x) = 0$.

By Theorem 3.1 we need to consider initial facet systems of the tuple of Newton polytopes. Observe that polynomials whose coefficients have all the same sign do not have real positive solutions. In order to find real positive solutions of the initial facet system a necessary requirement is that the signs of coefficients differ.

Definition 3.14. Given sparse polynomials $f_{ij} = \sum_{a \in \mathcal{A}_i} c_{ja} x^a \in \mathbb{R}[x_1, \dots, x_m]$, $i = 1, \dots, r$, $j = 1, \dots, k_i$, $\sum_i k_i = m = \dim(\text{conv}(\mathcal{A}_1) + \dots + \text{conv}(\mathcal{A}_r))$. A facet $F =$

$(F_1, \dots, F_r)$ of $(\text{conv}(\mathcal{A}_1), \dots, \text{conv}(\mathcal{A}_r))$ with inner normal ω is called alternating, if for all supports i for all F_i with $|F_i| \geq 2$ for all polynomials the signs of the coefficients of the monomials in $\omega(f_{ij}) = \sum_{a \in F_i} c_{ja} x^a$ are different. Analogously, a face with this sign property is called alternating.

The signed Newton polytopes of our chemical reaction system (5) have special properties. Often the dimension satisfies $\dim(\text{conv}(\mathcal{A}_i)) < m$. Often the support of one equation is the union of the supports of some other polynomials.

Observe that the support of the i th equation can be read off from the combined graph. The reaction complexes C_j being adjacent to S_i in the bipartite graph give monomials with negative coefficient. Each arrow $C_k \rightarrow C_j$ in the directed graph with an C_j being adjacent to S_i give a monomial with positive coefficient. So all monomials x^{η_j} above ($\eta_{ij} > 0$) have negative coefficient while all monomials below ($\eta_{kj} = 0$) have positive coefficient. For an example see Fig. 4.

The efficient computation of all alternating facets for a chemical reaction system (5) is left for future research.

Observe that each forward–backward reaction gives a pair of monomials with one negative and one positive coefficient. The polynomials contain either both monomials or none of them. Thus they are good candidates for the components F_i of alternating faces.

Lemma 3.15. *Consider a terminal strong connected component of the directed graph consisting of forward–backward reactions such that its complexes C_j , $j \in S$, satisfy conditions (D) and either (C^+) or (C_-) of Lemma 3.9. Then the tuple of Newton polytopes has an alternating face $F = (F_1, \dots, F_r)$.*

Proof. By Lemma 3.9, $\text{conv}(\eta_j, j \in S)$ is a face of the full Newton polytope. Thus for i with $\text{conv}(\eta_j, j \in S \cap \mathcal{A}_i) \neq \emptyset$, it is a face of the Newton polytope $\text{conv}(\mathcal{A}_i)$ for all $i \in A = \bigcup_{j \in S} T(x^{\eta_j})$. This defines a cone of directions ω attaining its minimum at the face of these $\text{conv}(\mathcal{A}_i)$. Since the terminal strong connected component consists of forward–backward reactions for each $i \in A$ there is a pair $C_j \leftrightarrow C_k$ with $j, k \in F_i$ whose coefficients of the associated monomials have different sign. Each ω in the cone attains its minimum over the other $i \notin A$ in at least one point. $\square$

Example 3.11 (continued). The two faces of the full Newton polytope give rise to two facets of the tuple of the Newton polytopes, see Fig. 6. $\omega_1 = (-2, 1, -1)$ defines the facet $(\{C_1, C_2\}, \{C_4, C_5\}, \{C_1, C_2\})$ while $\omega_2 = (2, 1, 1)$ defines the facet $(\{C_1, C_2\}, \{C_3, C_4\}, \{C_3, C_4\})$. Both are alternating and have positive solutions. This makes the results from one Newton polytope in Fig. 5 more precise. With one Newton polytope we saw that the curve is approaching two hypersurfaces in their limits. Distinguishing several Newton polytopes we obtain that the curve is approximating the curves $(\frac{k_{1,2}}{k_{2,1}}t^{-2}, \frac{k_{3,4}}{k_{6,5}}t, t^{-1})$ and $(\frac{k_{1,2}}{k_{2,1}}t^2, \frac{k_{3,4}}{k_{4,3}}t, t)$ on these hypersurfaces.

Besides terminal strong connected components also other structures of mass action kinetics lead to alternating faces. For example in [17] the authors discuss a chemical reaction with a so-called starter radical. This phenomenon is left for future research.

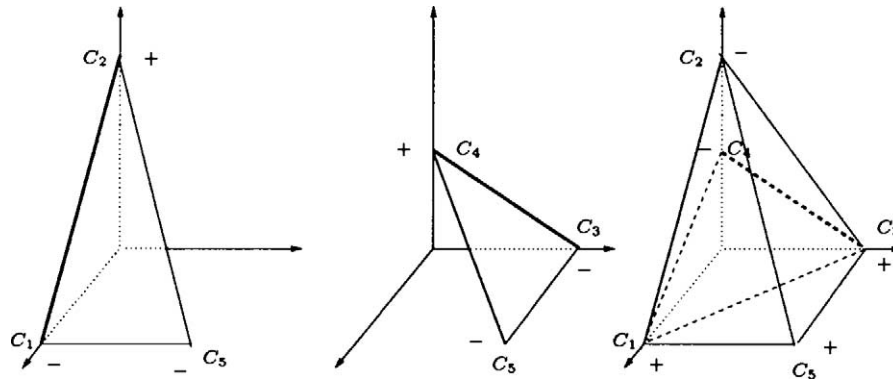


Fig. 6. Two alternating facets of the Newton polytopes of Example 3.11.

3.4. New results on positive solutions

The following theorem shows that the technique of alternating faces can be used to prove the existence of positive steady states for all parameter values.

Theorem 3.16. Assume a system with one conservation relation with $v_1 > 0$. Assume the solutions of $Y_s I_a I_K \Psi(x) = 0$ form a curve. Assume that one limit is $x = 0$ and one is non-negative infinity and that there are no other alternating cells with non-negative solutions. Then the system

$$Y_s I_a I_K \Psi(x) = 0, \quad v_1^t x - c_1 = 0$$

has at least one positive solution for all possible choices of $k_{ij} > 0$ and $c_1 > 0$.

Example 2.1 (continued). This example is an instance of the theorem. Computations show that there are two alternating facets of the tuple of Newton polytopes induced by E_1 and E_2 such that the initial facet system has a positive solution. The total number of facets is 29. But, all other facets are either not alternating or do not have a positive solution. Thus the real curve links the origin with positive infinity. There is at least one solution of system (5) and (6) for each positive value of the rate constants k_{ij} and for each positive value of the constant c_1 in the conservation relation.

Similar versions of Theorem 3.16 are easily formulated. The same principle is valid for several conservation relations, restrictions for the constants in the conservation relations, and for higher dimension of the real variety of $Y_s I_a I_K \Psi(x) = 0$.

Example 3.13 (continued). For the case of one Newton polytope we saw three facets whose initial face system has affine non-negative solutions. If we distinguish the supports of the single equations we find that the tuple of Newton polytopes has many alternating facets. Among these there are three facets corresponding to the connected components

whose initial facet system has a non-trivial solution. All other can be neglected. This polynomial system has for each positive value of the constants c_1, c_2, c_3 a positive solution.

These results agree with the fact that a system with mass action kinetics has a non-negative solution in each space $\{x \in (\mathbb{R}_{\geq 0})^m \mid v_i^t x - c_i = 0, i = 1, \dots\}$, provided they are bounded, see [28].

4. Multistationarity by Viro's method

In this section we clarify the importance of stoichiometric generators E_i and its confluent vectors $g_i = I_a E_i \neq 0$ for the presence of several positive solutions (multistationarity). We restrict to the case $Y = Y_s = Y_k \in (\mathbb{Z}_{\geq 0})^{m,n}$. Multistationarity happens in the limit $k_{ij} \rightarrow 0$ for some k_{ij} . In this limit the polynomial system (5) deforms to stoichiometric subgraph subsystems where Clarke's cone is generated by just one stoichiometric generator.

We start by discussing rescalings of the rate constants k_{ij} .

Lemma 4.1. *As in Section 2 we assume two graphs which define the system (5) and (6) with mass action kinetics. If $a \in \mathbb{R}_+^m$ is a solution of the unscaled system (5) and (6) then the scaled point $\tilde{a} = \text{diag}(\tau_1, \dots, \tau_m)a$ is a solution of the scaled system*

$$0 = Y I_a I_{\tilde{K}} \Psi(\tilde{x}), \quad 0 = \tilde{v}_i^t \tilde{x} - c_i$$

where $\tau_i \in \mathbb{R}_+, i = 1, \dots, m$, and

$$I_{\tilde{K}} = I_K \text{diag}\left(\frac{1}{\psi_1(\tau)}, \dots, \frac{1}{\psi_n(\tau)}\right), \quad \tilde{v}_i = \text{diag}\left(\frac{1}{\tau_1}, \dots, \frac{1}{\tau_m}\right)v_i.$$

Remark 4.2. $\tilde{v}_i$ is in general not orthogonal to $\text{im}(Y I_a I_{\tilde{K}})$. Thus the equivalent system is not of the form mass action kinetics as defined in Section 2.

We consider different rescalings of the rate constants k_{ij} which are equivalent to rescalings of x and v_i, c_i . The first type of rescaling is $k_{ij} \mapsto \tau k_{ij}$ for all arrows. Then Eq. (5) can be divided by τ and the system of polynomial equations does not change. The second rescaling $k_{ij} \mapsto \tau^{y_{ij}} k_{ij}$ for all arrows is equivalent to $x \mapsto (x_1, \dots, x_l/\tau, \dots, x_m)$ and $v_i \mapsto ((v_i)_1, \dots, (v_i)_l \cdot \tau, \dots, (v_i)_m)$. A combination of these rescalings is $k_{ij} \mapsto \tau^{|y_j|} k_{ij}$ for all arrows. It is equivalent to $x \mapsto (x_1/\tau, \dots, x_m/\tau)$ and $c_i \mapsto c_i/\tau$.

So we need to distinguish two different processes. One rescaling of k_{ij} which is equivalent to rescaling of x . The other one is given by choosing a representative of $\mathbb{C}^{l-1}/\text{im}(\mathbb{C}^R)$ where $R = \dim(\text{span}(y_2 - y_1, \dots, y_r - y_1))$ is the dimension of the lattice and r the number of reactant complexes.

According to this we suggest a change of coordinates from k to $k_{\text{new}} = (k_0, \tilde{k}_1, \dots, \tilde{k}_R, \hat{k}_1, \dots, \hat{k}_{l-R-1})$ by

$$\underline{k}_j = \prod_{i=1}^l (k_{\text{new}})_i^{b_{ij}}, \quad j = 1, \dots, l, \quad (k_{\text{new}})_j = \prod_{i=1}^l \underline{k}_i^{c_{ij}}, \quad j = 1, \dots, l, \quad (10)$$

where $B = (b_{ij}) \in \mathbb{Z}^{l,l}$ is invertible with inverse $B^{-1} = C = (c_{ij}) \in GL(l, \mathbb{Q})$. The first row of B consists of one's. The i th row of B ($i = 2, \dots, R+1$) is given by the degrees of monomials in the variable x_{i-1} (eventually one needs to change coordinates such that the variables correspond to a lattice basis). The columns $j = R+2, \dots, l$ of C are given by a module basis of the nullspace $\ker(\hat{Y}_L)$. Such transformation always exists and has many degrees of freedom. Then the deformed toric ideal $I_{\hat{Y}_L}$ depends on $\hat{k}_j$ only. Then a rescaling of x_i is equivalent to a rescaling of $\hat{k}_i$, $i = 1, \dots, R$. We refer to $\tilde{k}_i$, $i = 1, \dots, R$, as *toric rate constants*.

In the following lemma we are dealing with solutions of $I_a I_K \Psi(x) = 0$ which are obviously solutions of $Y I_a I_K \Psi(x) = 0$. By Lemma 5.1 in [21] the system $I_a I_K \Psi(x) = 0$ can only have positive solutions if each connected component of the directed graph is strongly connected, see also [24,30]. The content of the following lemma is basically Theorem 2.16 and Lemma 3.6 in [30], see also [24]. In order to understand the relevant mathematical structure we give an alternative algebraic proof based on the *Cayley trick* [22].

We first recall the idea of the Cayley trick for two polynomials in $\mathbb{C}[x_1, \dots, x_m]$: $f_1(x) = c_1 x^{a_1} + \dots + c_n x^{a_n}$, $f_2(x) = c_{n+1} x^{a_{n+1}} + \dots + c_L x^{a_L}$. A vector z in the nullspace of the coefficient matrix, i.e., $c_1 z_1 + \dots + c_n z_n = 0$, $c_{n+1} z_{n+1} + \dots + c_L z_L = 0$ is a representative z of a class $[z] \in \mathbb{P}^{n-1} \times \mathbb{P}^{L-1}$. It gives a solution x by solving the overdetermined system

$$\begin{aligned} z_1 &= x_0 x^{a_1}, & \frac{z_2}{z_1} &= x^{a_2 - a_1}, & \dots, & & \frac{z_n}{z_1} &= x^{a_n - a_1}, \\ z_{n+1} &= x_{-1} x^{a_{n+1}}, & \frac{z_{n+2}}{z_{n+1}} &= x^{a_{n+2} - a_{n+1}}, & \dots, & & \frac{z_L}{z_{n+1}} &= x^{a_L - a_{n+1}}. \end{aligned}$$

This has a solution if $z \in V(I_{\check{A}}) \subset \mathbb{P}^{n-1} \times \mathbb{P}^{L-1}$ where $\check{A}$ means a double lifting of A corresponding to two slack variables x_0, x_{-1} :

$$\check{A} = \begin{pmatrix} 1 & \dots & 1 & 0 & \dots & 0 \\ 0 & \dots & 0 & 1 & \dots & 1 \\ a_1 & \dots & a_n & a_{n+1} & \dots & a_L \end{pmatrix}.$$

Lemma 4.3. *Given a directed graph and bipartite graph as defined in Section 2. Assume that each connected component of the directed graph is strongly connected. There exists a set of values of the rate constants k_{ij} such that the following holds. For each set of positive values of the constants c_i in the conservation relations there exists a unique positive solution of $I_a I_K \Psi(x) = 0$ and $v_i^t x - c_i = 0$, $i = 1, \dots$.*

Proof. Define $z = I_K \Psi(x)$. If x is a positive solution of $I_a I_K \Psi(x) = 0$ then $z \in \ker(I_a) \cap (\mathbb{R}_+)^l$. That means we restrict Clarke's cone to the polyhedral subcone generated by positive circuits of the directed graph. Let these positive circuits denote by E_i , $i = 1, \dots, c$. Then each $z \in \ker(I_a) \cap (\mathbb{R}_+)^l$ is a convex combination of positive circuits, i.e., there exists $j_i > 0$ with $z = \sum_{i=1}^c j_i E_i$.

To each connected component we can associate a set of monomials and a projective space. z is a representative of a class in a product of these projective spaces. The polynomial system $I_a I_K \Psi(x) = 0$ is equivalent to a system consisting of special polynomials. Each polynomial has in its representation by monomials only monomials associated to one of the connected components. Consequently, we apply a modification of the Cayley trick.

For each connected component a slack variable x_{-i} , $i = 1, \dots, s$, is introduced, where s denotes the number of connected components. This corresponds to a lifting $\check{Y}_L \in \mathbb{Z}^{m+s,l}$ in the following way. Each column of Y_L is associated to a vertex ($\simeq$ complex $\simeq$ monomial) in one of the connected components. All columns of Y_L associated to the i th connected component get a lifting with entry 1 in the i th row and zero otherwise. The multihomogeneous ideal $I_{\check{Y}_L} \subset \mathbb{C}[w]$ has a variety in the product of projective spaces. Analogously, there is the multihomogeneous deformed toric ideal $I_{\check{Y}_L}^{\text{def}} \subset \mathbb{C}(\underline{k})[z]$ is defined. If $[z] \in V(I_{\check{Y}_L}^{\text{def}})$ and $z \in \ker(I_a) \cap (\mathbb{R}_+)^l$ then there exists associated positive solutions x .

Since $z > 0$ it is sufficient to look at a lattice basis ideal

$$I_C^{\text{def}} = \langle \{z^{g^+} - \underline{k}^g z^{g^-}, g \text{ element of a module basis of } \ker(\check{Y}_L)\} \rangle.$$

By Lemma 12.2 in [34], $I_C^{\text{def}} : (z_1 \cdots z_l)^\infty = I_{\check{Y}_L}^{\text{def}}$ and thus the positive parts of the varieties are equal, $V_+(I_{\check{Y}_L}^{\text{def}}) = V_+(I_C^{\text{def}})$.

The columns of Y_L consists of columns of Y . Some are repeated if the associated complex is an initial complex for several reactions. That means I_C^{def} includes linear polynomials. Since we demand $z = \sum_{i=1}^c j_i E_i \in V(I_C^{\text{def}})$ there are linear conditions for the rate constants k_{ij} . But this only relates rate constants inside a connected component and coefficients j_i of positive circuits of the same connected component. Besides this there is exactly one degree of freedom in z for each connected component. This agrees with the fact that $I_{\check{Y}_L}^{\text{def}}$ is multihomogeneous. We complete the linear polynomials to a generating system of I_C^{def} by a module basis $g_1, \dots, g_b$ of $\ker(\check{Y})$. This gives linear independent $\bar{g}_1, \dots, \bar{g}_b \in \ker(\check{Y}_L)$ and thus independent parameters $\underline{k}^{\bar{g}_1}, \dots, \underline{k}^{\bar{g}_b}$. Using $z \in V(I_C^{\text{def}})$ again gives special values of the rate constants by solving $z^{\bar{g}_1^+} - \underline{k}^{\bar{g}_1} z^{\bar{g}_1^-} = 0, \dots, z^{\bar{g}_b^+} - \underline{k}^{\bar{g}_b} z^{\bar{g}_b^-} = 0$ for the parameters. This proves the existence of positive solutions.

Whether there is one positive solution or infinitely many depends on the lattice associated to the Cayley trick which is the sum of the sublattices generated by the differences inside a connected component. If the lattice is $\mathbb{Z}^m$ then there is precisely one positive solution of $I_a I_K \Psi(x) = 0$. If the lattice is degenerate there are infinitely many solutions parameterized by free parameters and precisely one positive value for the determined variables. The rank of the lattice equals $\text{rank}(Y I_a)$.

The parameterization is transversal to each space $\{x \in \mathbb{R}_+^m \mid v_i^t x - c_i = 0, i = 1, \dots\}$ which implies uniqueness. $\square$

Remark 4.4. (i) Lemma 4.3 contains the deficiency-zero-theorem in [15,21] as a subcase. (ii) A positive solution x which satisfies $I_a I_K \Psi(x) = 0$ as in the lemma above is called complex balancing [24]. This name is justified by their existence and uniqueness and the fact that they are asymptotically stable [6]. (iii) The complex balancing solutions are unique

as solutions of $I_a I_K \Psi(x) = 0$, $v_i^t x - c_i = 0$ and unique as solutions of $Y I_a I_K \Psi(x) = 0$, $v_i^t x - c_i = 0$ which is shown in [24] using a Lyapunov function. (iv) In [24] a special subcase of complex balancing has another name. A solution x such that $k_{ij} \Psi_j(x) = k_{ji} \Psi_i(x)$ for all forward–backward reactions $C_i \leftrightarrow C_j$ is called detailed balancing. (v) In the complex balancing case the lattice associated to $\check{Y}$ is degenerate leading to free parameters. Two solutions a, c which just differ by values of this free parameters give rise to a vector $\ln(a) - \ln(c)$ which is orthogonal to the lattice which generates a vector space equal to $\text{im}(Y I_a)$. In [24] these two points are said to be quasi-thermostatic to each other.

In the proof a multiple lifting is introduced according to the connected components of the directed graph. We denote the multiple lifted matrices by $\check{Y}$ and $\check{Y}_L$ and $\check{Y}_e$. We call the polytope generated by the columns of $\check{Y}_L$, *Cayley-graph-polytope* in analogy to the Cayley polytope [22].

Now we turn the attention to various toric ideals. Besides the (multi)homogeneous toric ideals $I_{\check{Y}}, I_{\check{Y}_r}, I_{\check{Y}_L}, I_{\check{Y}_e}, I_{\check{Y}}, I_{\check{Y}_L}, I_{\check{Y}_e}$ we are dealing with (multi)homogeneous *deformed* toric ideals $I_{\check{Y}_L}^{\text{def}}, I_{\check{Y}_L}^{\text{def}}, I_{\check{Y}_e}^{\text{def}}, I_{\check{Y}_e}^{\text{def}}$. For the latter $I_{\check{Y}_e}^{\text{def}}, I_{\check{Y}_e}^{\text{def}}$ we need to introduce dummy parameters for the rate constants of pure product complexes.

$I_{\check{Y}_e}$ is related to minimal generators of Clarke's cone which are not positive circuits of the directed graph. If $\text{rank}(Y I_a) < \text{rank}(I_a)$ then such stoichiometric generators exist obviously. Each stoichiometric generator E_i of $\ker(Y I_a) \cap (\mathbb{R}_{\geq 0})^l$ yields a non-zero vector $I_a E_i$. By suitable rescaling the vector has integer entries which are coprime. Our notation is $g_i = \text{normalize}(I_a E_i)$. Following [16] we call g_i *confluent* vector. Let s be the number of stoichiometric generators. Of course $g_1, \dots, g_s$ may be linear dependent.

Lemma 4.5. *Given the directed graph and bipartite graph with matrices $Y_r, Y_L, Y_e, Y = Y_s = Y_k$ as in Section 2 we denote by $\check{Y}$ the lifting induced by the connected components of the directed graph.*

- (i) *The confluent vectors $g_1, \dots, g_s$ generate the nullspace of $\check{Y}$ as a vector space.*
- (ii) *The toric ideal $I_{\check{Y}}$ is multihomogeneous and contains binomials $w^{g_i^+} - w^{g_i^-}$, $i = 1, \dots, s$. Moreover,*

$$I_{\check{Y}} = \langle w^{g_1^+} - w^{g_1^-}, \dots, w^{g_s^+} - w^{g_s^-} \rangle : (w_1 \cdots w_n)^\infty.$$

- (iii) *Let $\bar{g}_i$ be a renumbering and filling with zeros of g_i such that $\bar{g}_i \in \ker(\check{Y}_e)$ for $i = 1, \dots, s$. Let I_D be the ideal generated by $z^{\bar{g}_i^+} - z^{\bar{g}_i^-}$, $i = 1, \dots, s$, and all linear polynomials $z_i - z_j$ corresponding to multiple initial complexes. Then*

$$I_{\check{Y}_e} = I_D : (z_1 \cdots z_d)^\infty,$$

where d is the number of columns of Y_e .

Proof. Since $\ker(Y I_a) = \ker(I_a) \oplus \text{span}(E_1, \dots, E_s)$ we have $\text{span}(g_1, \dots, g_s) \subseteq \ker(Y)$. In fact $\text{span}(g_1, \dots, g_s) \subseteq \ker(\check{Y})$ because $\ker(\check{Y}) \subset \text{im}(I_a)$ since the lifting $\check{Y}$ of Y is

induced by the directed graph. But I_a restricted to a complement of $\ker(I_a)$ is invertible. Thus $(I_a)^{-1}(\ker(\check{Y})) \subseteq \ker(YI_a)$. Consequently, $g_1, \dots, g_s$ generate $\ker(\check{Y})$. (ii) follows from (i) and Lemmas 4.1 and 12.2 in [34]. (iii) follows from elementary linear algebra and [34] again. $\square$

Remark 4.6. (i) For deficiency one $\dim(\ker(\check{Y})) = 1$ and its generator g_1 is a circuit of $\check{Y}$. (ii) If the directed graph has connected components which are all strongly connected then $Y_L = Y_e$ and Lemma 4.5(iii) gives $I_{\check{Y}_e}^{\text{def}} = I_D^{\text{def}} : (z_1 \cdots z_l)^\infty$.

In view of the Cayley trick we refine the new rate constants $k_{\text{new}} = (k_0, \tilde{k}, \hat{k})$. We consider the case that no complexes are pure product complexes which includes the case of Lemma 4.3 where all connected components of the directed graph are strongly connected. If the lattice generated by $y_2 - y_1, \dots, y_n - y_1$ has larger rank than the lattice associated to the Cayley-graph-polytope (which is generated by differences inside connected components) then we subdivide both tuples $\tilde{k}$ and $\hat{k}$.

It is possible to choose toric rate constants $\tilde{k}_1^C, \dots, \tilde{k}_c^C$ associated to the lattice of the Cayley-graph-polytope (with dimension c) and complete them to a full set of toric rate constants $\tilde{k}_{c+1}, \dots, \tilde{k}_R$ corresponding to the lattice.

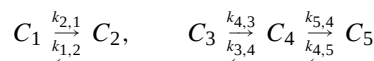
Obviously, $I_{\check{Y}} \subset I_{\hat{Y}}$ in $\mathbb{C}[w_1, \dots, w_n]$ and thus $I_{\check{Y}_L}^{\text{def}} \subset I_{\hat{Y}_L}^{\text{def}}$. With the help of Lemma 4.5 we subdivide the tuple $\hat{k}$. For a pair of reactions $C_i \rightarrow C_j$ and $C_i \rightarrow C_v$ we choose $k_{ijv} = k_{ji}/k_{vi}$. From the confluent vectors we choose a maximal linear independent set $g_1, \dots, g_t$ and choose

$$k_{g_i} = \underline{k}^{\tilde{g}_i} = \prod_{C_v \xrightarrow{j} C_\mu} (k_{\mu v})^{(\tilde{g}_i)_j}, \quad i = 1, \dots, t, \quad (11)$$

as parameters. We refer to them as *confluent rate constants*. The k_{g_i} and k_{ijv} are clearly some independent representatives of $\mathbb{C}^{l-1}/\mathbb{C}^R$. Some binomial generators in $I_{\check{Y}_L}^{\text{def}}$ which are not in $I_{\hat{Y}_L}^{\text{def}}$ give the remaining new rate constants.

Now we are ready to present an example for Lemma 4.3.

Example 4.7. This example is due to Edelstein and has been investigated before by Feinberg and in [21]. The reactions are



with stoichiometric matrix

$$Y = \begin{pmatrix} 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \quad \text{yielding} \quad \begin{aligned} 0 &= k_{2,1}x_1 - k_{1,2}x_1^2 - k_{4,3}x_1x_2 + k_{3,4}x_3, \\ 0 &= -k_{4,3}x_1x_2 + (k_{3,4} + k_{5,4})x_3 - k_{4,5}x_2, \\ 0 &= k_{4,3}x_1x_2 + (-k_{3,4} - k_{5,4})x_3 + k_{4,5}x_2. \end{aligned}$$

The conservation relation is $x_2 + x_3 - c_1 = 0$. The cone is generated by three positive circuits and two stoichiometric generators which are linear dependent.

$$\begin{aligned} E_1 &= (1, 1, 0, 0, 0, 0), & E_2 &= (0, 0, 1, 1, 0, 0), & E_3 &= (0, 0, 0, 0, 1, 1), \\ E_4 &= (1, 0, 1, 0, 1, 0), & E_5 &= (0, 1, 0, 1, 0, 1). \end{aligned}$$

Then $g_4 := I_a E_4 = (-1, 1, -1, 0, 1)$ and $I_a E_5 = -g_4$. The confluent vector gives the Gröbner basis of $I_{\tilde{Y}} = \langle w^{g_4^+} - w^{g_4^-} \rangle = \langle w_2 w_5 - w_1 w_3 \rangle$. All complexes are reaction complexes, but the fourth complex gives rise to two reactions. Thus the deformed toric variety contains a linear relation. The two homogeneous toric ideals $I_{\tilde{Y}_L}$ and $I_{\hat{Y}_L}$ are equal and thus

$$I_{\tilde{Y}_L}^{\text{def}} = I_{\hat{Y}_L}^{\text{def}} = \langle z_4 k_{5,4} - z_5 k_{3,4}, z_6 z_2 k_{2,1} k_{4,3} - z_1 z_3 k_{4,5} k_{1,2} \rangle.$$

This suggests to use $\hat{k}_{435} = \underline{k}^{e_{45}} = \frac{k_{3,4}}{k_{5,4}}$ and $k_{g_4} = \bar{k}^{\bar{g}_4} = \frac{k_{4,5} k_{1,2}}{k_{2,1} k_{4,3}}$ as new parameters (toric rate constants), where $\underline{k} = (k_{2,1}, k_{1,2}, k_{4,3}, k_{3,4}, k_{5,4}, k_{4,5})$ and $e_{45} = (0, 0, 0, 1, -1, 0)$ and $\bar{g}_4 = (-1, 1, -1, 0, 0, 1)$. They can be read off from the Cayley trick. Each $[z] \in \mathbb{P}^1 \times \mathbb{P}^3$ which is in $V(I_{\tilde{Y}_L}^{\text{def}})$ such that z is in the cone generated by E_1, E_2, E_3 gives rise to complex balancing solutions by solving

$$\begin{aligned} z_1 &= k_{2,1} x_0 x_1, & \frac{z_3}{z_1} &= \frac{k_{4,3} x_{-1}}{k_{2,1} x_0} x_2, & \frac{z_5}{z_3} &= \frac{k_{5,4}}{k_{4,3}} \frac{x_3}{x_1 x_2}, & \frac{z_6}{z_3} &= \frac{k_{4,5}}{k_{4,3}} \frac{1}{x_1}, \\ & & \frac{z_2}{z_1} &= \frac{k_{1,2}}{k_{2,1}} x_1, & \frac{z_4}{z_3} &= \frac{k_{3,4}}{k_{4,3}} \frac{x_3}{x_1 x_2}. \end{aligned} \quad (12)$$

Only Eqs. (12) are important for the Cayley trick. They suggest the change of coordinates $x_1 = w_1, x_2 = w_3, x_3 = w_1 w_2 w_3$ or equivalently $w_1 = x_1, w_2 = x_3 / (x_1 x_2), w_3 = x_2$. That means Eqs. (12) determine w_1, w_2 and leave w_3 arbitrary. The lattice associated to the Cayley trick has dimension two while the whole lattice has dimension three. So the new coordinates are $k_{\text{new}} = (k_0, \tilde{k}_1^C, \tilde{k}_2^C, \tilde{k}_3, \tilde{k}_{435}, k_{g_4})$. The ansatz for a change of coordinates with B and C gives an affine linear space of solutions of dimension 8. We choose

$$\begin{aligned} k_{2,1} &= \frac{k_0}{\tilde{k}_2^C}, & k_{1,2} &= \frac{k_0 \tilde{k}_1^C}{\tilde{k}_2^C}, & k_{4,3} &= \frac{k_0 \tilde{k}_3}{\tilde{k}_2^C}, \\ k_{3,4} &= k_0 \tilde{k}_3, & k_{5,4} &= \frac{k_0 \tilde{k}_3}{\hat{k}_{435}}, & k_{4,5} &= \frac{k_0 \tilde{k}_3 k_{g_4}}{\tilde{k}_1^C \tilde{k}_2^C} \end{aligned}$$

or equivalently

$$\begin{aligned} k_0 &= \frac{k_{2,1} k_{3,4}}{k_{4,3}}, & \tilde{k}_1^C &= \frac{k_{1,2}}{k_{2,1}}, & \tilde{k}_2^C &= \frac{k_{3,4}}{k_{4,3}}, & \tilde{k}_3 &= \frac{k_{4,3}}{k_{2,1}}, \\ \hat{k}_{435} &= \frac{k_{3,4}}{k_{5,4}}, & k_{g_4} &= \frac{k_{4,5} k_{1,2}}{k_{2,1} k_{4,3}}. \end{aligned}$$

Then

$$I_{\hat{Y}_L}^{\text{def}} = \langle z_4 - \hat{k}_{435} z_5, z_6 z_2 - k_{g_4} z_1 z_3 \rangle.$$

The toric variety is independent of $\tilde{k}_1^C, \tilde{k}_2^C, \tilde{k}_3$ which are the rescalings of k_{ij} inherited from the torus action on x . For each $z \in V(I_{\hat{Y}_L}^{\text{def}}) \subset \mathbb{P}^1 \times \mathbb{P}^3$ we need to solve $z_2/z_1 = \tilde{k}_1^C w_1$ and $z_4/z_3 = \tilde{k}_2^C w_2$ from (12). Remember for complex balancing solutions $z = \sum_{i=1}^5 j_i E_i$ with $j_4 = j_5 = 0$ or equivalently $z_1 = z_2, z_3 = z_4, z_5 = z_6$. Then $z \in V(I_{\hat{Y}_L}^{\text{def}})$ implies $\hat{k}_{435} = j_2/j_3$ and $k_{g_4} \hat{k}_{435} = 1$. This gives the family of complex balancing solutions

$$x = \left(\frac{1}{\tilde{k}_1^C}, w_3, \frac{w_3}{\tilde{k}_1^C \tilde{k}_2^C} \right)$$

which is obviously transversal to the affine spaces $\{x \mid x_2 + x_3 - c_1 = 0\}$ in $(\mathbb{R}_+)^3$. Observe that the complex balancing solutions are independent of $\tilde{k}_3$.

We still need to discuss how we change coordinates for the case that the directed graph has pure product complexes. For each pure product complex C_i we suggest to introduce a slack constant $k_{i,i}$. Instead of studying the intersection of Clarke's cone with $V(I_{\hat{Y}_L}^{\text{def}})$ we amend the cone by new coordinates z_{l+j} . Then we intersect Clarke's cone with the variety of $I_{\hat{Y}_e}^{\text{def}}$. Since the confluent vectors g_i give rise to vectors $\tilde{g}_i \in \ker(\check{Y}_e)$ we can still use the confluent rate constants k_{g_i} . Eventually, the number of toric rate constants increases. Only choices of new rate constants k_{new} are relevant which corresponds to $k_{i,i} = 0$ for all pure product complexes. An example will be given below.

Before describing the transition to the limit we start with the situation arising in the limit itself.

Proposition 4.8. Assume a system (5) with mass action kinetics such that the following conditions are satisfied.

- (i) The directed graph is a forest of trees and each connected component contains only one strong connected component (here a sink).
- (ii) The system has deficiency one: $\text{rank}(Y I_a) - \text{rank}(I_a) = 1$ and there exists a generator $E \in (\mathbb{R}_+)^l$ of $\ker(Y I_a)$ which is positive.
- (iii) $\hat{Y}_r$ has maximal rank r , the number of reactant complexes.

Then there exist values of the constants c_i in the conservation relations such that the space $\{x \in \mathbb{R}_+^l \mid v_i^t x - c_i = 0, i = 1, \dots, m - \text{rank}(Y I_a)\}$ contains at least one positive solution of $Y I_a I_K \Psi(x) = 0$. If $r < m$ then the real positive variety of $Y I_a I_K \Psi(x) = 0$ has higher dimension and is parameterized by $m - r$ parameters. Assume the directed graph consists of more than one connected component and $y_1, \dots, y_r$ are the columns of Y corresponding to reaction complexes. Assume there exists a vector $\mu \in \text{span}(y_2 - y_1, \dots, y_r - y_1)^\perp$ such that

- (iv) The vector μ is not orthogonal to $\text{im}(Y I_a)$.
- (v) There exists a positive solution x of $Y I_a I_K \Psi(x) = 0$ such that $\text{diag}(x)\mu \in \text{im}(Y I_a)$.

Then in a generic situation there exist positive values for c_i such that $\{x \in \mathbb{R}_+^l \mid v_i^t x - c_i = 0, i = 1, \dots\}$ contains at least two positive solutions of the system $Y I_a I_K \Psi(x) = 0$.

Proof. We study the intersection of the convex polyhedral cone $\ker(Y I_a) \cap (\mathbb{R}_{\leq 0})^l$ with the deformed toric variety $V(I_{\hat{Y}_r}^{\text{def}})$. By (i) there are no minimal generators of the cone which are positive circuits of the directed graph. By (ii) the cone is generated by one stoichiometric generator E . In order to solve $E = I_K \Psi(x) \cdot x_0$ we observe that $\text{rank}(I_a) = \text{rank}(I_a I_K)$ if each connected component contains just one strong connected component. By (i) this guarantees that $E = I_K \Psi(x) x_0$ is equivalent to a binomial system $x_0 x^{y_1} = b_1, \dots, x_0 x^{y_r} = b_r$ where the $y_1, \dots, y_r$ correspond to reactant complexes and the b_i are positive numbers depending on the values of k_{ij} and the components of E . The toric variety $V(I_{\hat{Y}_r})$ is equivalent to the projective space $\mathbb{P}^{r-1}$ by (iii). Solving the binomial system by Hermite normal form $U \hat{Y}_r = H$ we introduce new coordinates

$$x_0 = w_0^{u_{00}} \cdots w_m^{u_{m0}}, \quad \dots, \quad x_m = w_0^{u_{0m}} \cdots w_m^{u_{mm}}, \quad (13)$$

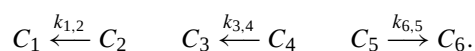
where $U = (u_{ij})_{i,j=0,\dots,m}$ is unimodular. For the first set of new coordinates $w_1, \dots, w_r$ there is precisely one positive solution of the binomial system. The coordinates $w_{r+1}, \dots, w_m$ not appearing in the system parameterize the variety. Choosing for these coordinates values equal to 1 and substituting $x = x(w)$ into the conservation relations we find values of c_i . If $m > r$, then there exist unrestricted variables $w_i, i = r+1, \dots, m$. In (13) the submatrix $U^s = (u_{ij})_{j=1,\dots,m, i=r+1,\dots,m}$ determines the dependence on $w_{r+1}, \dots, w_m$. The rows $\mu_i = \text{row}_i(U^s)$ are orthogonal to the lattice $y_2 - y_1, \dots, y_r - y_1$ and define a tangent to the variety, i.e.,

$$\frac{dx}{dw_i} = \text{diag}(x) \mu_i \frac{1}{w_i}, \quad i = r+1, \dots, m.$$

If for some particular values of $w_{r+1}, \dots, w_m$ and $\mu = \sum_{i=r+1}^m \mu_i (1/w_i)$ the vector $\text{diag}(x(w_{r+1}, \dots, w_m)) \mu$ is an element of $\text{im}(Y I_a)$ then the variety is touching the space $\{x \mid v_i^t x - c_i, i = 1, \dots\}$ for special values of c_i . This happens if $m - r \geq m - \text{rank}(Y I_a)$ and there is a $\mu \in \text{span}(\mu_{r+1}, \dots, \mu_m) = \text{span}(y_2 - y_1, \dots, y_r - y_1)^\perp$ which is not orthogonal to $\text{im}(Y I_a)$ (condition (iv)) and $\text{diag}(x)$ is rescaling μ to an element of $\text{im}(Y I_a)$ (condition (v)). Condition (iv) is necessary for condition (v). Under a genericity condition on higher derivatives there is multistationarity. $\square$

Remark 4.9. A variation of the lemma still states the existence of a solution. There (i) is weakened to the structure of a forest of trees only. To (ii) we add: for each pair of arrows $C_j \rightarrow C_i, C_j \rightarrow C_p$ with indices μ, ν the stoichiometric generator satisfies $k_{ij} E_\nu - k_{pj} E_\mu = 0$.

Example 3.11 (continued). We consider the subsystem given by the stoichiometric generator $E_3 = (0, 1, 0, 1, 1)$. The directed graph is



Then the polynomial subsystem is

$$\begin{aligned} 0 &= k_{1,2}x_3^2 - k_{6,5}x_1x_2, \\ 0 &= k_{3,4}x_3 - k_{6,5}x_1x_2, \\ 0 &= -2k_{1,2}x_3^2 - k_{3,4}x_3 + 3k_{6,5}x_1x_2, \\ 0 &= 2x_1 + x_2 + x_3 - c_1 \end{aligned} \quad \text{with } Y_r = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix} \text{ and } E = (1, 1, 1).$$

A solution is given by intersecting the nullspace of the coefficient matrix (here $\text{span}(E)$) with the homogeneous toric variety which is $\mathbb{P}^2$. Solving $k_{1,2}x_0x_3^2 = 1$, and $k_{3,4}x_0x_3 = 1$ and $k_{6,5}x_0x_1x_2 = 1$ yields

$$(x_1, x_2, x_3) = \left(\frac{k_{3,4}^2}{k_{1,2}k_{6,5}w}, w, \frac{k_{3,4}}{k_{1,2}} \right), \quad (14)$$

where w is an arbitrary number, since the lattice is degenerate.

For $w = k_{3,4}\sqrt{2}/\sqrt{k_{1,2}k_{6,5}}$ the tangent to the curve $\frac{dx}{dw} = (-1/2, 1, 0) \in \text{im}(YI_a)$ is orthogonal to the conservation vector v_1 . For

$$c_1 = \frac{(k_{6,5} + 2\sqrt{2}\sqrt{k_{1,2}k_{6,5}})k_{3,4}}{k_{1,2}k_{6,5}}$$

the curve is touching the space $\{x \in \mathbb{R}_+^3 \mid 2x_1 + x_2 + x_3 = c_1\}$, while for larger values of c_1 the curve intersects in two positive solutions $x, \tilde{x}$ corresponding to w and $\tilde{w}$. Then

$$\mu = (\ln(x_1/\tilde{x}_1), \ln(x_2/\tilde{x}_2), \ln(x_3/\tilde{x}_3)) = \ln(w/\tilde{w}) \cdot (-1, 1, 0),$$

may be defined which is not parallel to v_1 , but orthogonal to the Newton polytope generated by y_2, y_4, y_5 , see Fig. 10.

In order to study multistationarity we need more definitions.

Definition 4.10 [34]. Given $A = (a_1, \dots, a_n) \subset \mathbb{R}^{m,n}$ of rank m . A triangulation Δ of $\text{conv}(a_1, \dots, a_n)$ is a set of m -simplices $\mathcal{S}_i = \{a_{i_0}, \dots, a_{i_m}\}$ such that $\text{conv}(a_1, \dots, a_n) = \bigcup_i \text{conv}(\mathcal{S}_i)$ and $\text{conv}(\mathcal{S}_i) \cap \text{conv}(\mathcal{S}_j) = \text{conv}(\mathcal{S}_i \cap \mathcal{S}_j)$, $\forall i \neq j$. A triangulation is called regular and denoted Δ_ω if it is induced by a lifting, i.e., there are values $\omega_1, \dots, \omega_n$ such that $\text{conv}(\mathcal{S}_i)$ are the projected lower facets of $\text{conv}(A^\omega)$ where

$$A^\omega = \begin{pmatrix} \omega_1 & \dots & \omega_n \\ a_1 & \dots & a_n \end{pmatrix}.$$

Definition 4.11 [34]. For a triangulation Δ of $a_1, \dots, a_n$ the *Stanley–Reisner ideal* is defined as

$$I_\Delta = \left\langle \left\{ \prod_{j \in \sigma} z_j \mid \sigma \subset \{1, \dots, n\} \text{ such that } a_j, j \in \sigma \text{ do not form a face of } \Delta \right\} \right\rangle.$$

Recall that for a given ideal $I \subset k[x_1, \dots, x_n]$ almost all term orders can be represented by a vector $\omega \in \mathbb{R}^n$. Given an ideal, all term orders $\omega \in \mathbb{R}^n$ with the same reduced Gröbner basis form a cone. There are finitely many cones forming the *Gröbner fan*. A result in [34] states that each reduced Gröbner basis of I_A yields a regular triangulation. Moreover, $\sqrt{\text{in}_\omega(I_{\hat{A}})} = I_{\Delta}$ for a term order and lifting ω . Circuits of $\hat{A}$ are important since they distinguish two regular triangulations. All lifting vectors ω of A giving the same regular triangulation form a cone. The set of these cones is the *secondary fan*. The Gröbner fan of a toric ideal $I_{\hat{A}}$ is a refinement of the secondary fan. More about this can be found in [34, Chapters 1–3, 8], [3,22,31].

Now we will be interested in regular triangulations of the Cayley-graph-polytope, of the Newton polytope as well as of the polytope which is the convex hull of the columns of Y_L , or of Y_e .

The Gröbner fans of toric ideals are special. Since I_A is weighted homogeneous with respect to each row we can restrict to consider term orders $\omega \in \ker(A)$.

Lemma 4.12. *Assume a vector space basis of $\ker(\check{Y})$ is given by confluent vectors $g_1, \dots, g_t$. Each term order of $I_{\check{Y}}$ is represented by a vector $\omega = \omega_1 g_1 + \dots + \omega_t g_t$. There exists a vector of signs $v_1, \dots, v_t \in \{+, -\}$ and a regular triangulation $\check{\Delta}$ of $\check{Y}$ such that*

$$\langle w^{\gamma_1}, \dots, w^{\gamma_t} \rangle \subset \text{in}_\omega(I_{\check{Y}}) \quad \text{and} \quad \sqrt{\text{in}_\omega(I_{\check{Y}})} = I_{\check{\Delta}}$$

with $\gamma_1 = g^{v_1}, \dots, \gamma_t = g^{v_t}$ being the positive or negative parts of the confluent vectors. In particular, there exists a term order ω with $\langle w^{g_1^+}, \dots, w^{g_t^+} \rangle \subset \text{in}_\omega(I_{\check{Y}})$.

Proof. Since toric ideals are multihomogeneous we may restrict to vectors $\omega \in \ker(\check{Y}) = \text{span}(g_1, \dots, g_t)$ to represent term orders. For each confluent vector g_i the polynomial $w^{g_i^+} - w^{g_i^-}$ is an element of $I_{\check{Y}}$ by Lemma 4.5. If $\omega^t g_i^+ > \omega^t g_i^-$ then $lt(w^{g_i^+} - w^{g_i^-}) = w^{g_i^+}$ and thus $w^{g_i^+} \in \text{in}_\omega(I_{\check{Y}})$ else $w^{g_i^-} \in \text{in}_\omega(I_{\check{Y}})$. The existence of the regular triangulation follows from [34]. $\square$

Each confluent vector g_i defines a hyperplane $\{w \in \mathbb{R}^n \mid w^t g_i = 0\}$ which divides the space $\mathbb{R}^n$ into halfspaces. The intersections of all these halfspaces are convex polyhedral cones which form a fan. The lemma above states that the Gröbner fan of $I_{\check{Y}}$ is a refinement of this fan.

In general the confluent vectors g_i are not primitive and thus the associated binomials are in general not elements of the universal Gröbner basis.

Lemma 4.13. *Assume a vector space basis of $\ker(\check{Y})$ is given by confluent vectors $g_1, \dots, g_t$. The associated elements of $\ker(\check{Y}_e)$ are denoted by $\bar{g}_1, \dots, \bar{g}_t$ and completed by differences $e_{ij} = e_i - e_j$ of unit vectors to form a vector space basis of $\ker(\check{Y}_e)$. For each*

$$\omega = \omega_1 \bar{g}_1 + \dots + \omega_t \bar{g}_t + \sum_{C_k \xrightarrow{i} C_\nu, C_k \xrightarrow{j} C_\mu} \omega_{ij} e_{ij},$$

representing a term order for $I_{\check{Y}_e}$, the following holds.

- (i) For $i = 1, \dots, s$ there is a sign v such that $w^{\bar{g}_i^v} \in \text{in}_\omega(I_{\check{Y}_e})$.
- (ii) For each pair of reactions $C_k \xrightarrow{i} C_v, C_k \xrightarrow{j} C_\mu$ either $w_i \in \text{in}_\omega(I_{\check{Y}_e})$ or $w_j \in \text{in}_\omega(I_{\check{Y}_e})$.
- (iii) There exists a regular triangulation $\check{\Delta}$ of $\check{Y}_e$ such that its Stanley–Reisner ideal $I_{\check{\Delta}}$ is the radical of the initial ideal $\text{in}_\omega(I_{\check{Y}_e})$.

Proof. This is a corollary of the previous lemma since the columns of Y and Y_e are equal besides that in Y_e some columns are repeated if a complex is the initial complex for several reactions. $\square$

The term order $\omega = \sum_{i=1}^t \omega_i g_i$ is also a lifting vector. If $\omega = g_i$ is generic, then the lifting defines a regular triangulation, otherwise it defines a cell complex.

Lemma 4.14. Let $g_1, \dots, g_s$ denote the confluent vectors. Then for $i = 1, \dots, s$:

- (i) The cell complex of $\check{Y}$ associated to g_i includes a face given by $\text{supp}(g_i^-)$.
- (ii) The cell complex of $\check{Y}_e$ associated to $\bar{g}_i$ includes a face given by $\text{supp}(\bar{g}_i^-)$.

Proof. The lifting of y_j is zero, if $j \in \text{supp}(g_i)$. It is positive, if $j \in \text{supp}(g_i^+)$, and negative for $j \in \text{supp}(g_i^-)$. Since the cardinality of $\text{supp}(g_i^-)$ is less than or equal to the rank of $\check{Y}$, the lifted vector $y_j, j \in \text{supp}(g_i^-)$, generates a face of the lower hull of the polytope generated by $\check{Y}^{g_i}$. By projection this gives a face of the regular triangulation which proves (i). Analogously, (ii) is proved. $\square$

Of course, regular triangulations/cell complexes defined by lifting vectors sufficiently close to g_i , still have the face associated to $\text{supp}(g_i^-)$. The lemma shows that each confluent vector g_i , is a representative of a cone in the secondary fan of $\check{Y}$. If $\text{in}_\omega(I_{\check{Y}_e}^{\text{def}})$ is square-free, then we know more about the regular triangulation $\check{\Delta}_\omega$. For $j = 1, \dots, s$ the following holds. If $g_j^+ \in \text{in}_\omega(I_{\check{Y}_e}^{\text{def}})$, then $\text{supp}(g_j^-)$ defines a face of $\check{\Delta}_\omega$. If $g_j^- \in \text{in}_\omega(I_{\check{Y}_e}^{\text{def}})$, then $\text{supp}(g_j^+)$ defines a face.

For each cone in the Gröbner fan we investigate a toric deformation.

Proposition 4.15. Assume a term order $\omega \in \mathbb{R}^d$ for $I_{\check{Y}_e}^{\text{def}}$ which is a linear combination of $\bar{g}_1, \dots, \bar{g}_t, e_{ij}$ and assume the matrix B of the change of rate constants (10). The components of $\Omega = (B^t)^{-1}\omega$ corresponding to $\bar{g}_i, e_{ij}$ are denoted by ω_i, ω_{ij} . Then the one-parameter family of new rate constants

$$k_{g_i} = \varepsilon^{\omega_i}, \quad i = 1, \dots, t, \quad \hat{k}_{kv\mu} = \varepsilon^{\omega_{ij}} \quad \text{for } C_k \xrightarrow{i} C_v, C_k \xrightarrow{j} C_\mu$$

defines a flat deformation from $I_{\check{Y}_e}^{\text{def}}$ to the monomial ideal $\text{in}_\omega(I_{\check{Y}_e}^{\text{def}})$.

Proof. Each ω which represents a term order gives a reduced Gröbner basis of $I_{\check{Y}_e}^{\text{def}}$ which consists of binomials in $Q(\hat{k}_{kv\mu}, k_{gi})[z]$, since the ideal is toric. Each element in the reduced Gröbner basis is of the form

$$\underline{z^{\alpha^+}} - \underline{k^\alpha} z^{\alpha^-} = \underline{z^{\alpha^+}} - \left(\prod_{C_k \xrightarrow{i} C_v, C_k \xrightarrow{j} C_\mu} (\hat{k}_{kv\mu})^{\alpha_{ij}} \prod_{i=1}^t k_{gi}^{\alpha_i} \right) z^{\alpha^-}$$

with $\alpha \in \ker(\check{Y}_e) = \text{span}(e_{ij}, \bar{g}_1, \dots, \bar{g}_t)$ and thus $\alpha = Ca = \sum_{ij} \alpha_{ij} e_{ij} + \sum_{i=1}^t \alpha_i \bar{g}_i$. The special choice of a one-parameter family gives

$$\underline{z^{\alpha^+}} - \varepsilon^{\omega^t \alpha} z^{\alpha^-}$$

because $\sum_i \alpha_i \omega_i + \sum_{ij} \alpha_{ij} \omega_{ij} = a^t \Omega = (Ba)^t (B^t)^{-1} \omega = \alpha^t \omega$. On the other hand, we know that z^{α^+} is the leading term and thus $\omega^t \alpha > 0$. Then $\lim_{\varepsilon \rightarrow 0} (z^{\alpha^+} - \varepsilon^{\omega^t \alpha} z^{\alpha^-}) = z^{\alpha^+}$. This proves the deformation to the leading monomial ideal. $\square$

Lemma 4.16. *The projective variety $V(\text{in}_\omega(I_{\check{Y}_e}^{\text{def}})) \subseteq \mathbb{P}^{d-1}$ is a collection of projective subvarieties which are isomorphic to projective spaces (of dimension $\text{rank}(\check{Y}) - \#\text{components} - 1$). For each subvariety there is a simplex in the associated regular triangulation $\check{\Delta}_\omega$ of $\check{Y}_e$ and vice versa. For each projective subvariety the intersection with the (amended) Clarke's cone $\text{cone}(E_1, \dots)$ is a subcone generated by some of the minimal generators E_i .*

The subcones define non-negative solutions of system (5) in a general way. Each projective space corresponds to a subcone and it corresponds to a subsystem of (5), since components of z are zero.

This is a generalization of Viro's method. In the original Viro method a homotopy parameter t is artificially introduced. The limit $t \rightarrow 0$ deforms the homogeneous toric variety into projective spaces. This also defines a regular triangulation with a subsystem for each simplex. The solutions of the subsystems are deformed to solutions of the original system [22,32,33,35].

Definition 4.17. For a simplex in a regular triangulation $\check{\Delta}$ we define a subsystem of $Y I_a I_K \Psi(x) = 0$ in the following way. Denote the columns of $Y I_a$ by $(Y I_a)_j$, $j = 1, \dots, d$. For the set of indices $J \subseteq \{1, \dots, d\}$ of a simplex we define

$$\sum_{j \in J} (Y I_a)_j (I_K \Psi(x))_j$$

and perform the change of coordinates (10) and substitute $k_{gi} = 1$ and $\hat{k}_{ijv} = 1$ for all i, j .

Theorem 4.18. *Assume the setting above, in particular the flat deformation, the regular triangulation $\check{\Delta}$ of $\check{Y}_e$ and the subsystems associated to the simplices of $\check{\Delta}$. The set $\{x \in$*

$(\mathbb{R}_{\geq 0})^m \mid \exists \hat{k}_j > 0 \text{ with } Y I_a I_K \Psi(x) = 0\}$ deforms by $\varepsilon \rightarrow 0$ via the flat deformation to the generalized solutions of the subsystems for all $\hat{k}_j > 0$. In particular, for fixed values of $\hat{k}_j, \tilde{k}_{c+1}, \dots, \tilde{k}_R$ the intersection with spaces $\{x \mid v_i^t x - c_i = 0, i = 1, \dots\}$ deforms to generalized solutions of the subsystems which also satisfy the conservation relations $v_i^t x - c_i = 0, i = 1, \dots$.

Observe that finding the positive solutions of a subsystem is related to Clarke's cone. The positive solutions of a subsystem of a simplex with index set J are determined by those minimal generators E_i with $\text{supp}(E_i) \subseteq J$. If we use coordinates w associated to the lattices (instead of x) and keep $\hat{k}_j$ arbitrary, then we only determine some of the w_i , namely those which are associated to the lattice of the Cayley-graph-polytope.

When one solves for positive solutions for fixed values of $\hat{k}_j, \tilde{k}_{c+1}, \dots, \tilde{k}_R$ one still needs to consider the ideal $I_{\check{Y}_e}^{\text{def}}$ additional to $I_{\check{Y}_e}^{\text{def}} \subseteq I_{\check{Y}_e}^{\text{def}}$ as in Theorem 4.18. If a cell of a regular triangulation of $\hat{Y}_e$ shows multistationarity, then there is multistationarity for small ε by continuity.

Example 4.7 (continued). Since C_4 is an initial complex for two reactions the matrix $Y = (y_1, y_2, y_3, y_4, y_5)$ is expanded to $Y_e = Y_L = (y_1, y_2, y_3, y_4, y_4, y_5)$. Term orders of

$$I_{\check{Y}_e}^{\text{def}} = I_{\check{Y}_e}^{\text{def}} = I_{\check{Y}_L}^{\text{def}} = \langle z_4 - \hat{k}_{435} z_5, z_6 z_2 - k_{g_4} z_1 z_3 \rangle$$

can be restricted to $\text{span}(e_{45}, \bar{g}_4)$. There are four different reduced Gröbner bases.

- (1) For $\omega = e_{45} + \bar{g}_4$ we deform the toric ideal with $\hat{k}_{435} = \varepsilon, k_{g_4} = \varepsilon$ and $\varepsilon \rightarrow 0$ to $\langle z_4, z_6 z_2 \rangle = \langle z_2, z_4 \rangle \cap \langle z_4, z_6 \rangle$. This corresponds to a regular triangulation with simplices $\{z_1, z_3, z_5, z_6\}$ and $\{z_1, z_2, z_3, z_5\}$. The intersections of the two subvarieties $V(\langle z_2, z_4 \rangle), V(\langle z_4, z_6 \rangle)$ with Clarke's cone are the boundary subcones $\text{cone}(E_3, E_4)$ and $\text{cone}(E_1, E_4)$.
- (2) For $\omega = -e_{45} - \bar{g}_4$ we deform the toric ideal with $\hat{k}_{435} = \varepsilon^{-1}, k_{g_4} = \varepsilon^{-1}$ and $\varepsilon \rightarrow 0$ to $\langle z_5, z_1 z_3 \rangle = \langle z_1, z_5 \rangle \cap \langle z_3, z_5 \rangle$. This corresponds to a regular triangulation with simplices $\{z_2, z_3, z_4, z_6\}$ and $\{z_1, z_2, z_4, z_6\}$. The intersections of the two subvarieties with Clarke's cone are the boundary subcones $\text{cone}(E_1, E_5)$ and $\text{cone}(E_2, E_5)$.
- (3) For $\omega = e_{45} - \bar{g}_4$ the deformation gives $\langle z_4, z_1 z_3 \rangle = \langle z_1, z_4 \rangle \cap \langle z_3, z_4 \rangle$. The subcones are generated by E_3 and by E_1, E_3 , respectively.
- (4) For $\omega = -e_{45} + \bar{g}_4$ the deformation gives $\langle z_5, z_2 z_6 \rangle = \langle z_2, z_5 \rangle \cap \langle z_5, z_6 \rangle$. The subcones are generated by E_2 , as well as by E_1, E_2 .

The cases (3) and (4) are the two extreme subcases of the case of complex balancing solutions studied before.

The cases (1) and (2) use the regular triangulations of the Cayley-graph-polytope of $\check{Y}_e$. In this example the projections of the two regular triangulations of $\check{Y}_e$ give regular triangulations of the Newton polytope $\text{conv}(y_1, \dots, y_5)$ which are presented in Fig. 7. The inner face $\{z_1, z_3, z_5\}$ of the first triangulation corresponds to the support of E_4 and includes the support of $\bar{g}_4^-$. The inner face $\{z_2, z_4, z_6\}$ of the second triangulation corresponds to

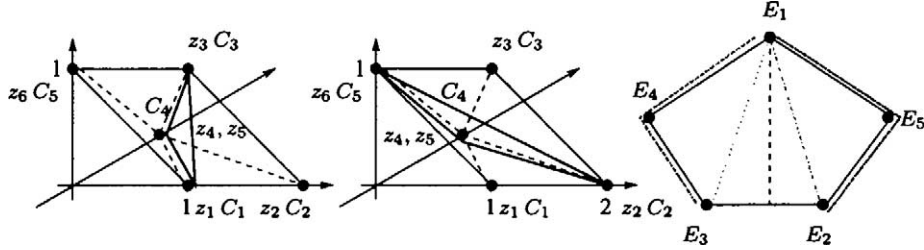


Fig. 7. Edelman example: two regular triangulations (left), Clarke's cone (right). The middle line is the intersection with the deformed toric variety for $\hat{k}_{435}k_{g4} = 1$ which corresponds to complex balancing solutions. The four line collections corresponds to the four flat deformations.

$$\begin{array}{lll} C_1 \rightleftharpoons C_2, & C_3 \rightarrow C_4 \rightarrow C_5 & C_1 \rightleftharpoons C_2, & C_3 \leftarrow C_4 \leftarrow C_5 \\ C_1 \rightarrow C_2, & C_3 \rightarrow C_4 \rightleftharpoons C_5 & C_1 \leftarrow C_2, & C_3 \rightleftharpoons C_4 \leftarrow C_5 \end{array}$$

Fig. 8. Edelman example: four subnetworks associated to the four cells of the two regular triangulations in Fig. 7.

the support of E_5 and includes the support of $\bar{g}_5^-$. The inner normal of the first inner face is $\mu = (1, 0, 1)$. For the second it is $\mu = (1, 2, 2)$. Both normals are not parallel to $v_1 = (0, 1, 1)$. But condition (v) of Proposition 4.8 is not satisfied. There is no multistationarity for the stoichiometric subsystems. Nevertheless there is multistationarity in the limit of case (1).

The set $\{x \in (R_+)^3 \mid \exists \tilde{k}_3 > 0 \text{ with } Y I_a I_K \Psi(x) = 0\}$ deforms by Theorem 4.18. Since $I_{\tilde{Y}} = I_{\tilde{Y}}$, we can immediately discuss the case for fixed $\tilde{k}_3$ and solve simpler equations associated to the simplices of the triangulation. In case (1), for the cell with indices $(1, 2, 3, 5)$, the simpler equations are with $z = j_1 E_1 + j_4 E_4 = (j_1 + j_4, j_1, j_4, 0, j_4, 0)$

$$\frac{z_2}{z_1} = \frac{j_1}{j_1 + j_4} = \tilde{k}_1^C w_1, \quad \frac{z_5}{z_3} = 1 = \tilde{k}_2^C w_2, \quad \frac{z_3}{z_1} = \frac{j_4}{j_1 + j_4} = \tilde{k}_3^C w_3$$

yielding

$$x = (w_1, w_3, w_1 w_2 w_3) = \left(\frac{j_1}{(j_1 + j_4) \tilde{k}_1^C}, \frac{j_4}{(j_1 + j_4) \tilde{k}_3^C}, \frac{j_1 j_4}{(j_1 + j_4)^2 \tilde{k}_1^C \tilde{k}_2^C \tilde{k}_3^C} \right).$$

For $j_1 = 0, j_4 = 1$ the curve is touching a space $\{x \mid x_2 + x_3 = c_1\}$ which is also true in a neighborhood, see Fig. 9. For the second cell $(1, 3, 5, 6)$ and $z = j_3 E_3 + j_4 E_4 = (j_4, 0, j_4, 0, j_3 + j_4, j_3)$ we solve

$$\frac{z_5}{z_3} = \frac{j_3 + j_4}{j_4} = \tilde{k}_2^C w_2, \quad \frac{z_6}{z_3} = \frac{j_3}{j_4} = \frac{1}{\tilde{k}_1^C} \frac{1}{w_1}, \quad \frac{z_3}{z_1} = 1 = \tilde{k}_3^C w_3,$$

yielding

$$x = \left(\frac{j_4}{j_3 \tilde{k}_1^C}, \frac{1}{\tilde{k}_3^C}, \frac{(j_3 + j_4)}{j_3 \tilde{k}_1^C \tilde{k}_2^C \tilde{k}_3^C} \right).$$

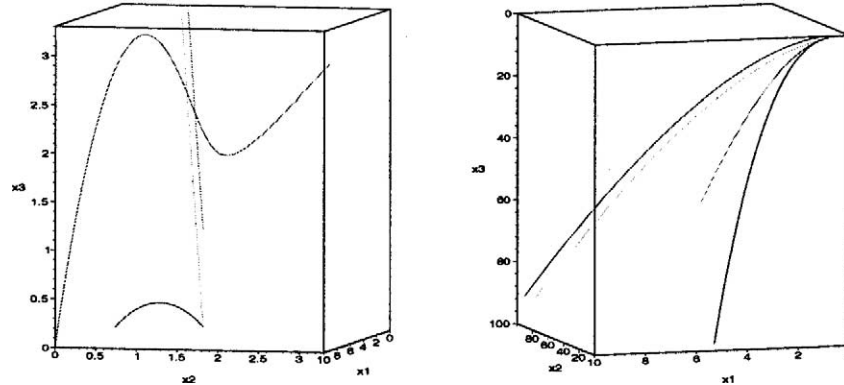


Fig. 9. Left: solutions for $k_{g4} = 0.1$, for both cells and for E_4 . Right: solutions for $k_{g4} = 5$ of cell (E_1, E_5) , E_5 , the system and cell (E_2, E_5) (from left to right).

When j_1 is small x is dominated by j_4 for both cells. The dependence is given by $\mu = (1, 0, 1)$, the normal on the joint face. This agrees with Bernstein's second theorem.

For the first cell $(1, 2, 4, 6)$ in case (2) we have for $z = j_1 E_1 + j_5 E_5 = (j_1, j_1 + j_5, 0, j_5, 0, j_5)$

$$\frac{z_2}{z_1} = \frac{j_1 + j_5}{j_1} = \tilde{k}_1^C w_1, \quad \frac{z_6}{z_4} = 1 = \frac{1}{\tilde{k}_1^C \tilde{k}_2^C} \frac{1}{w_1 w_2}, \quad \frac{z_4}{z_1} = \frac{j_5}{j_1} = \tilde{k}_2^C \tilde{k}_3 w_2 w_3$$

giving $w_1 = \frac{j_1 + j_5}{j_1 \tilde{k}_1^C}$, $w_2 = \frac{j_1}{(j_1 + j_5) \tilde{k}_2^C}$, $w_3 = \frac{j_5(j_1 + j_5)}{j_1^2 \tilde{k}_3}$, and yielding altogether

$$x = \left(\frac{j_1 + j_5}{j_1 \tilde{k}_1^C}, \frac{j_5(j_1 + j_5)}{j_1^2 \tilde{k}_3}, \frac{j_5(j_1 + j_5)}{j_1^2 \tilde{k}_1^C \tilde{k}_2^C \tilde{k}_3} \right).$$

For cell $(2, 3, 4, 6)$ we have with $z = j_2 E_2 + j_5 E_5 = (0, j_5, j_2, j_2 + j_5, 0, j_5)$

$$\frac{z_4}{z_3} = \frac{j_2 + j_5}{j_2} = \tilde{k}_2^C w_2, \quad \frac{z_6}{z_3} = \frac{j_5}{j_2} = \frac{1}{\tilde{k}_1^C w_1}, \quad \frac{z_3}{z_2} = \frac{j_2}{j_5} = \frac{\tilde{k}_3}{\tilde{k}_1^C} \frac{w_3}{w_1}$$

yielding

$$x = \left(\frac{j_2}{j_5 \tilde{k}_1^C}, \frac{j_2^2}{j_5^2 \tilde{k}_3}, \frac{j_2^2(j_2 + j_5)}{j_5^3 \tilde{k}_1^C \tilde{k}_2^C \tilde{k}_3} \right).$$

For both cells, for small values of j_1, j_2 the solutions depend on j_5 like $\mu = (1, 2, 2)$, the normal to the inner face of the regular triangulation. This agrees with Bernstein's second theorem.

Figure 9 shows the real curves for $k_0 = \tilde{k}_1^C = \tilde{k}_2^C = \tilde{k}_3 = 1$ fixed and for two pairs of values of $k_{g4}, \hat{k}_{435}$, e.g., $k_{g4} = \hat{k}_{435} = 0.1$ and $k_{g4} = \hat{k}_{435} = 5$.

Example 3.11 (continued). The lattice associated to the Cayley-graph-polytope is generated by $y_2 - y_1, y_4 - y_3, y_6 - y_5$ and has dimension 2. The lattice generated by $y_2 - y_1, y_3 - y_1, y_4 - y_1, y_5 - y_1$ has dimension 3. According to this, we choose the change of coordinates

$$x_1 = \frac{w_1}{w_2^2 w_3^3}, \quad x_2 = \frac{w_1}{w_2^2 w_3}, \quad x_3 = \frac{w_1}{w_2 w_3}, \quad \text{and}$$

$$w_1 = \frac{x_3^2}{x_1}, \quad w_2 = \frac{x_3}{x_2}, \quad w_3 = \frac{x_2}{x_1}.$$

For the change of coordinates on the rate constants we introduce a dummy parameter $k_{6,6}$ since C_6 is a pure product complex. According to the lattices and the confluent vector $g_3 = I_a E_3 = I_a(0, 1, 0, 1, 1) = (1, -1, 1, -1, -1, 1)$ we choose

$$k_0 = \frac{k_{3,4}^2}{k_{1,2}}, \quad \tilde{k}_1^C = \frac{k_{1,2}}{k_{2,1}}, \quad \tilde{k}_2^C = \frac{k_{3,4}}{k_{4,3}}, \quad \tilde{k}_3 = \frac{k_{4,3}}{k_{2,1}},$$

$$k_g = \frac{k_{2,1} k_{4,3} k_{6,6}}{k_{1,2} k_{3,4} k_{5,6}}, \quad \hat{k} = \frac{k_{3,4}^2 k_{6,5}}{k_{2,1} k_{1,2} k_{4,3}},$$

$$k_{2,1} = \frac{k_0 \tilde{k}_1^C}{(\tilde{k}_2^C)^2 \tilde{k}_3^2}, \quad k_{1,2} = \frac{k_0 (\tilde{k}_1^C)^2}{(\tilde{k}_2^C)^2 (\tilde{k}_3)^2}, \quad k_{4,3} = \frac{k_0 \tilde{k}_1^C}{(\tilde{k}_2^C)^2 \tilde{k}_3},$$

$$k_{3,4} = \frac{k_0 \tilde{k}_1^C}{\tilde{k}_2^C \tilde{k}_3}, \quad k_{6,5} = \frac{k_0 (\tilde{k}_1^C)^2 \hat{k}}{(\tilde{k}_2^C)^4 (\tilde{k}_3)^3}, \quad k_{6,6} = \frac{k_0 (\tilde{k}_1^C)^3 k_g \hat{k}}{(\tilde{k}_2^C)^3 (\tilde{k}_3)^3}.$$

In these coordinates a Gröbner basis of $I_{\hat{Y}_e}^{\text{def}}$ with respect to a weighted order is

$$\underline{z_4 z_6} - k_g \hat{k} z_2^2, \quad \hat{k} \underline{z_1 z_2 z_3} - z_4^2 z_5, \quad \underline{z_1 z_3 z_6} - k_g z_2 z_4 z_5.$$

The ideal includes $I_{\hat{Y}_e}^{\text{def}} = \langle z_1 z_3 z_6 - k_g z_2 z_4 z_5 \rangle$ and $I_{\hat{Y}_L}^{\text{def}} = \langle z_4^2 z_5 - \hat{k} z_1 z_2 z_3 \rangle$.

The limit $1/k_g \rightarrow 0$ deforms $I_{\hat{Y}_e}^{\text{def}}$ to a monomial ideal with primary decomposition $\langle z_2 \rangle \cap \langle z_4 \rangle \cap \langle z_5 \rangle$. This corresponds to a regular triangulation of the Cayley-graph-polytope with three tetrahedra. The intersection of $V(\langle z_2 \rangle)$ ($V(\langle z_4 \rangle)$, $V(\langle z_5 \rangle)$), respectively) with the amended Clarke's cone $\text{cone}(E_1, E_2, E_3)$ are $\text{cone}(E_2, E_4)$, $(\text{cone}(E_1, E_4), \text{cone}(E_1, E_2, E_4))$, respectively. The subsystems have only complex balancing solutions. There is no multistationarity in the limit.

The limit $k_g \rightarrow 0$ deforms $I_{\hat{Y}_e}^{\text{def}}$ to the monomial ideal $(z_1 z_3 z_6)$. The triangulation of the Cayley-graph-polytope consists of three tetrahedra. The intersections of the varieties with the cone are $\text{cone}(E_2, E_3, E_4)$, $\text{cone}(E_1, E_3, E_4)$, $\text{cone}(E_1, E_2, E_3)$. The set $\{x \in \mathbb{R}_+^3 \mid \exists \hat{k} > 0 \text{ with } Y I_a I_K \Psi(x) = 0\}$ deforms to three sets of solutions of the subsystems. Here, only w_1, w_2 are determined.

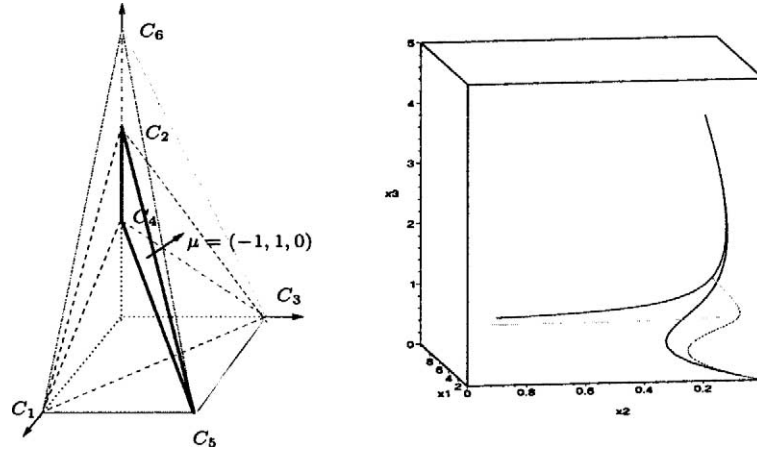


Fig. 10. Left: a triangulation with inner face induced by E_3 . Right: solution curves of the system, for two subsystems and of stoichiometric subsystem of E_3 .

For fixed $\hat{k}$ we need to study a deformation $I_{\hat{Y}_e}^{\text{def}}$. Term orders are elements of $\ker(\hat{Y}_e)$ which includes $\ker(\check{Y}_e)$. In this examples $w = g_3$ is already a term order and $\text{in}_{g_3}(I_{\hat{Y}_e}^{\text{def}} = \langle z_4 z_6, z_1 z_2 z_3, z_1 z_3 z_6 \rangle$ with primary decomposition

$$\langle z_1, z_4 \rangle \cap \langle z_1, z_6 \rangle \cap \langle z_3, z_4 \rangle \cap \langle z_3, z_6 \rangle \cap \langle z_2, z_6 \rangle.$$

Associated to this there is a regular triangulation of $\text{conv}(y_1, y_1, y_3, y_4, y_5, y_6)$ with five tetrahedra, see Fig. 10. The intersections of the five protective spaces with the cone are $\text{cone}(E_4)$, $\text{cone}(E_2, E_3)$, $\text{cone}(E_1, E_4)$, $\text{cone}(E_1, E_3)$, $\text{cone}(E_2)$. The second and fourth cell are interesting and the positive solutions of its subsystems are plotted in Fig. 10, together with the solution curve of the full system and the solutions of the subsystem associated to E_3 (14). The plot is for $\tilde{k}_1^C = \tilde{k}_2^C = \tilde{k}_3 = 1$, $\hat{k} = 10$. Whether multistationarity arises still depends on the choice of B .

The inner face (see Fig. 10) gives rise to multistationarity. Being an inner facet implies the inequalities $\mu^t y_2 > \mu^t y_1$, $\mu^t y_4 < \mu^t y_3$ which are part of the system of equalities and inequalities in [16].

Theorem 4.18 shows that the simplices of a regular triangulation of the Cayley-graph-polytope are the source of multistationarity. When the stoichiometric generators define inner faces of the regular triangulation, they can be considered as source of multistationarity as suggested by an example in [10]. The subsystem associated to an inner face given by a stoichiometric generator has positive solutions and thus guarantees that two neighboring simplices have positive solutions. But we need to restrict to those stoichiometric generators whose support involve several connected components. This theorem enables also an explanation of Femberg's work on multistationarity [16].

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